EXAM PACK

EEE475

RF and Microwave Engineering



Exam Pack

Session 2019–20 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2023*

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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2019–20

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PAST PAPER SOLUTIONS

EEE475

RF and Microwave Engineering

Question Paper Solutions

Session 2019–20

4th Year 1st Semester Final Examination, 2023

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	26 marks	26 marks

EACH ANSWER



Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Group A · **Answer any two of the following questions**				
Q1	(a) Directivity, Effective Height and Beam Area	9	16 min	☐
	(b) Antenna Aperture and Directivity	6	11 min	
	(c) Passive Remote Sensing	5	9 min	
	(d) Beamwidths, Radiated Power and Directivity of a Given Pattern	5	9 min	
Q2	(a) Constant-r Circles in the Reflection Coefficient Plane	6	11 min	☐
	(b) Properties of the r-Circles and x-Circles	6	11 min	
	(c) TE and TM Modes in a Circular Waveguide	3	5 min	
	(d) Attenuation and Phase Constants of a Lossy Line	10	18 min	
Q3	(a) S-Parameters and the Three-Port S-Matrix	6	11 min	☐
	(b) Condition for Constant Gain of an Amplifier	5	9 min	
	(c) Signal Flow Graph and Input Impedance of a Loaded Line	5	9 min	
	(d) Maximum Source Gain and Constant-Gain Circles of a FET	9	16 min	
Group B · **Answer any two of the following questions**				
Q4	(a) Short Notes: Reflection Coefficient, SWR, RF Hazards and Cavity Resonators	8	14 min	☐
	(b) Quarter-Wave Matching	4	7 min	
	(c) The Radar Range Equation	7	13 min	
	(d) Short-Circuited Stub Matching of a 50 Ohm Line	6	11 min	
Q5	(a) Wave Impedance, Cutoff Frequency and Phase Constant of TM Waves	6	11 min	☐
	(b) Forward Voltage Gain of a High-Pass Filter	6	11 min	
	(c) Uniform Linear Array	7	13 min	
	(d) Near Field and Far Field of an Antenna	7	13 min	
Q6	(a) Effect of Radio Frequency on Inductor and Capacitor Reactance	6	11 min	☐
	(b) Characteristic Impedance and Reflection Coefficient from the Line Equations	9	16 min	
	(c) Input Impedance of Open- and Short-Circuited Lines	4	7 min	
	(d) Load Impedance from SWR and Voltage-Minimum Measurements	7	13 min	

Formulas follow Pozar, Microwave Engineering (lines, waveguides, S-parameters, radar), Ludwig and Bretchko, RF Circuit Design (RF components, amplifier gain circles) and Kraus, Antennas (antennas, arrays, remote sensing).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2023
Course Code: EEE 475, Session: 2019-2020
Course Title: RF and Microwave Engineering

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Group A

*[Answer any **two** of the following questions]*

1. (a) Write short notes on the following topics
   i. Directivity
   ii. Effective Height
   iii. Beam Area 9

   (b) What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna. 6

   (c) Describe how passive remote sensing is used to detect object. 5

   (d) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$. 5

2. (a) Show that the normalized resistances represent circles in reflection coefficient plane. 6

   (b) Write the properties of r-circles and x-circles of a Smith chart. 6

   (c) Describe how TE mode is different from TM mode in a circular waveguide. 3

   (d) The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit [*sic*: should be short-circuit] is replaced by a load impedance $Z_L = 37 + j45$ Ω. 10

3. (a) Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. 6

   (b) Explain the condition for constant gain for amplifier. 5

   (c) For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation. 5

   (d) A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S. 9

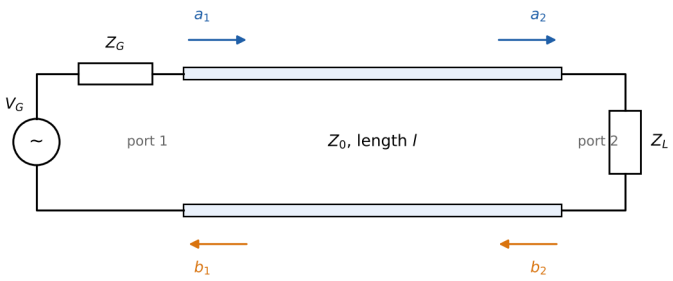


Figure 1 of the paper (redrawn)

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Group B

*[Answer any **two** of the following questions]*

4. (a) Write short notes on the following topic

   i. Reflection coefficient
   ii. Standing wave ratio
   iii. RF radiation hazards and safety standard
   iv. Microwave cavity resonators 8

   (b) Show how quarter wavelength line is used for matching. 4

   (c) Explain how the RADAR range equation is used to detect object. 7

   (d) A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line. 6

5. (a) Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave. 6

   (b) Find the forward voltage gain of a high-pass filter. 6

   (c) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. 7

   (d) Define near field and far field for an antenna. 7

6. (a) Explain how radio frequency affects the inductor and capacitor reactance. 6

   (b) From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient. 9

   (c) Briefly describe input impedance for open circuit termination and short circuit termination. 4

   (d) The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L. 7

Some useful Equations

$G_s = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$
$g_s = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}\,(1 - \lvert S_{11}\rvert^2)$
$d_{gi} = \dfrac{g_i S_{11}^*}{1 - \lvert S_{ii}\rvert^2(1 - g_i)}$
$r_{gi} = \dfrac{\sqrt{1 - g_i}\,(1 - \lvert S_{ii}\rvert^2)}{1 - \lvert S_{ii}\rvert^2(1 - g_i)}$

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1. (a) Write short notes on the following topics

   i. Directivity
   ii. Effective Height
   iii. Beam Area Marks: 9

   (b) What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna. Marks: 6

   (c) Describe how passive remote sensing is used to detect object. Marks: 5

   (d) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$. Marks: 5

1.(a) Write short notes on the following topics i. Directivity ii. Effective Height iii. Beam Area Marks: 9

1. Theory

1. What is a radiation pattern? (Core Idea)

An antenna does not send its power equally in all directions. If we walk around it at a fixed, large distance and measure the power per unit solid angle (the **radiation intensity** U, in W/sr), we get a curve with a big "lobe" in some direction and little or nothing elsewhere. Divide U by its largest value and we get the **normalized power pattern** $P_n(\theta, \phi) = U/U_{max}$, which is 1 at the peak.

A direction in 3-D space is given by two angles, θ (from the z axis) and ϕ (around it). A small patch of directions is a **solid angle** $d\Omega = \sin\theta\, d\theta\, d\phi$, measured in **steradians** (sr). The whole sphere is 4π sr.

2. Three ways to say "how concentrated is the beam?"

- **Beam area:** pretend all the power goes out uniformly inside a cone and nothing outside. How big must that cone be?
- **Directivity:** how many times stronger is the peak than the average over the sphere?
- **Effective height:** how long a straight wire with a *uniform* current would behave like this antenna? This one describes the antenna as a receiver of voltage.

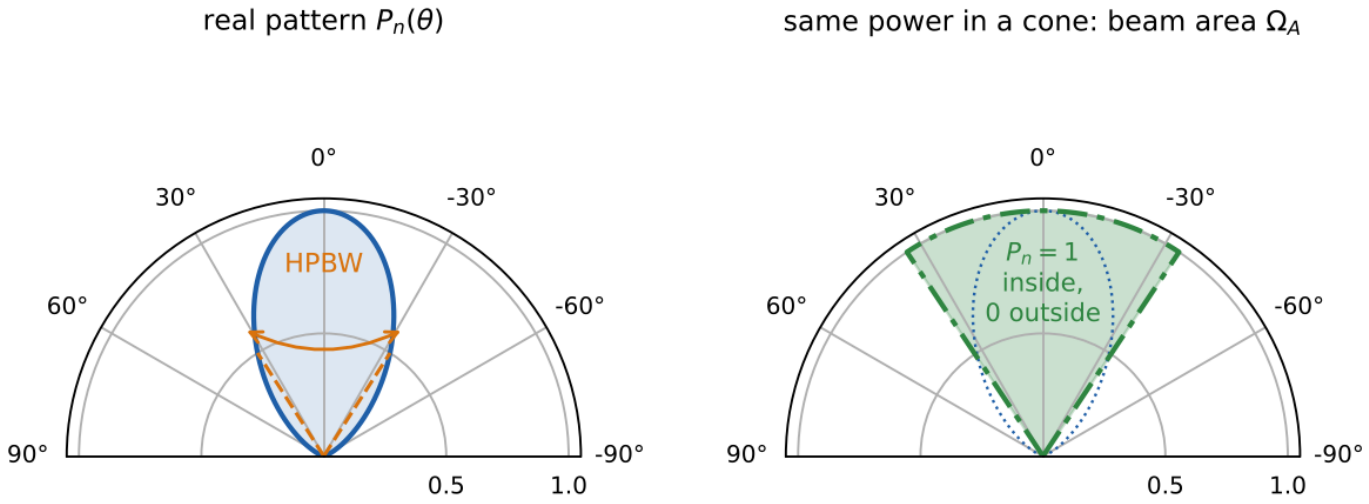


Figure 1: Beam area: the real pattern (left) and the cone of the same total power (right)

Reading the figure. On the left, the solid blue curve is a real beam; the dashed orange lines mark its

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half-power points (HPBW). On the right, the dash-dot green sector has height 1 inside and 0 outside, and holds the same total power as the real beam (dotted blue). Its angular size is the beam area.

2. Answer

i. Directivity

Definition: The directivity D of an antenna is the ratio of its maximum radiation intensity to the average radiation intensity over all directions:

$$D = \frac{U_{max}}{U_{av}} = \frac{U_{max}}{P_{rad}/4\pi} = \frac{4\pi U_{max}}{P_{rad}}$$

where:

- U_{max} = largest radiation intensity (W/sr), P_{rad} = total radiated power (W).

Key Points:

- An isotropic antenna (same in every direction) has $D = 1$ (0 dBi); every real antenna has $D > 1$.
- In terms of the beam area: $D = \dfrac{4\pi}{\Omega_A}$. A narrower beam gives a larger directivity.
- Estimate from the two half-power beamwidths (in degrees): $D \approx \dfrac{41\,253}{\theta_{HP}\,\phi_{HP}}$.
- The **gain** includes losses: $G = k\,D$, where $k \leq 1$ is the radiation efficiency.
- Examples: short dipole $D = 1.5$; half-wave dipole $D = 1.64$ (2.15 dBi).

ii. Effective Height

Definition: The effective height h_e (in metres) of a receiving antenna is the ratio of the open-circuit voltage at its terminals to the incident electric field (for a matched polarization):

$$h_e = \frac{V_{oc}}{E}$$

As a transmitter it is the length of an equivalent antenna that carries a *uniform* current equal to the terminal current I_0 and radiates the same far field. The two current distributions must have the same area:

$$h_e = \frac{1}{I_0}\int_{-l/2}^{l/2} I(z)\,dz$$

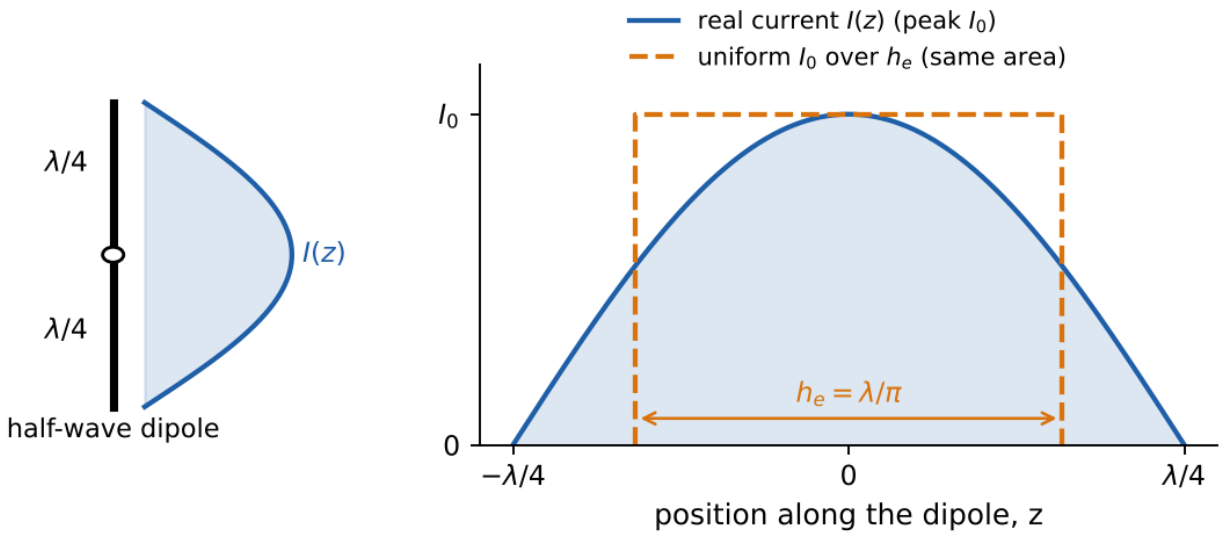


Figure 2: Effective height of a half-wave dipole

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Reading the figure. Left: a half-wave dipole with its sinusoidal current (solid blue), largest at the feed. Right: the same current plotted along the dipole; the dashed orange rectangle carries the uniform current I_0 over a length h_e and has the same area under it.

Key Points:

- Short dipole with a uniform current: $h_e = l$. With the usual triangular current: $h_e = l/2$.
- Half-wave dipole: $h_e = \dfrac{1}{I_0}\int_{-\lambda/4}^{\lambda/4} I_0 \cos\beta z\, dz = \dfrac{\lambda}{\pi} \approx 0.318\lambda$.
- It links to the effective aperture: $A_e = \dfrac{h_e^2\, Z_0}{4R_r}$ ($Z_0 = 377\ \Omega$, R_r = radiation resistance). For the half-wave dipole this gives $0.13\lambda^2$.

iii. Beam Area

Definition: The beam area (beam solid angle) Ω_A is the solid angle through which all the power would flow if the radiation intensity stayed at its maximum value inside it and were zero outside:

$$\Omega_A = \int_0^{2\pi}\int_0^{\pi} P_n(\theta,\phi)\, \sin\theta\, d\theta\, d\phi \quad \text{sr}$$

Key Points:

- It is measured in steradians or square degrees ($1\ \text{sr} = (180/\pi)^2 = 3283$ square degrees).
- **Approximation:** $\Omega_A \approx \theta_{HP}\, \phi_{HP}$, the product of the two half-power beamwidths (in radians).
- It sets the directivity, $D = 4\pi/\Omega_A$, and the effective aperture, $A_e\, \Omega_A = \lambda^2$.
- An isotropic source has $\Omega_A = 4\pi$ sr.

3. Keyword Explanation

1. Radiation intensity

Power radiated per unit solid angle, $U = r^2 S$ (W/sr), where S is the power density at distance r in the far field. It does not depend on r.

2. Steradian

The solid angle that cuts an area r^2 from a sphere of radius r. The whole sphere is 4π sr.

4. Notes and Remarks

- **Why** $D = 4\pi/\Omega_A$**:** $P_{rad} = \oint U\, d\Omega = U_{max} \oint P_n\, d\Omega = U_{max}\Omega_A$, so $D = 4\pi U_{max}/P_{rad} = 4\pi/\Omega_A$.
- Effective height and effective aperture both describe the same antenna as a receiver: h_e gives the voltage, A_e the power. Effective aperture is explained in 1(b); the effective height of a dipole is also used in 6(d) of the 2020–21 paper.

1.(b) What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna. Marks: 6

1. Theory

1. An antenna as a "collector" (Core Idea)

A receiving antenna sits in a passing wave that carries a power density S (W/m^2). It takes some power P_r out of the wave and delivers it to its load. We can ask: *how big a window, held square to the wave, would let through the same power?* That window area is the antenna's **aperture**. For a dish or a horn it is close to the physical mouth; for a thin wire it is much bigger than the wire itself.

2. Why a bigger aperture gives a sharper beam

A wave leaving an opening of width a spreads out by an angle of about λ/a radians (diffraction, as with light through a slit). So an opening many wavelengths wide sends out a narrow beam. A narrower beam puts the power into a smaller solid angle, which is exactly what a higher directivity means (1(a)).

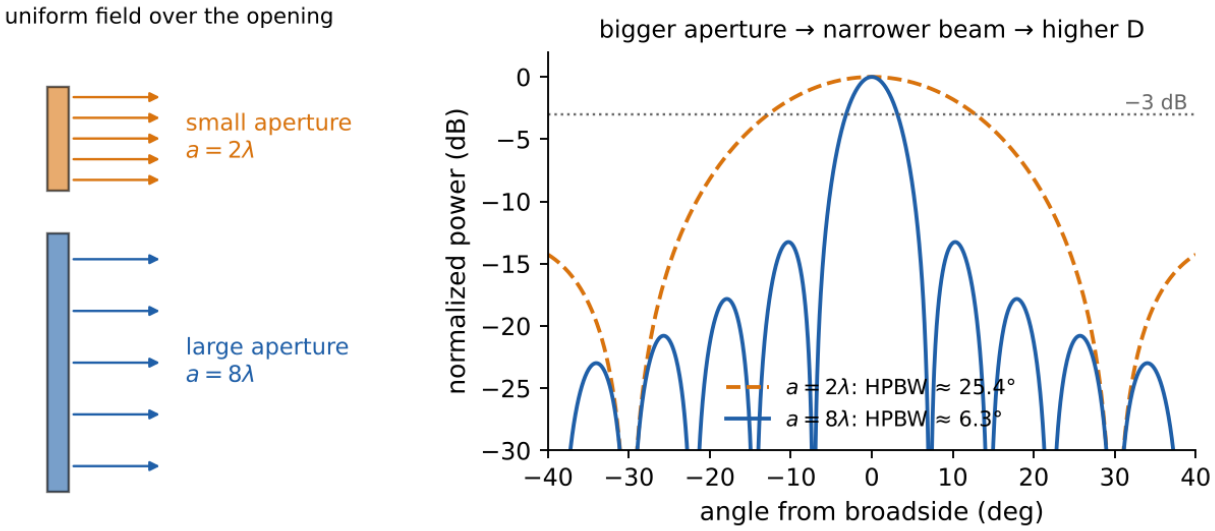


Figure 3: Bigger aperture, narrower beam

Reading the figure. Left: two openings with a uniform field: the short solid orange bar, 2λ wide, and the long solid blue bar, 8λ wide. Right: their far-field patterns in dB. The dashed orange pattern of the small aperture has a half-power beamwidth of about 25°; the solid blue one of the large aperture only about 6°.

2. Answer

Definition: The **effective aperture** A_e of an antenna is the ratio of the power it delivers to a matched load to the power density of the incident wave:

$$A_e = \frac{P_r}{S} \quad \mathrm{m}^2$$

The **physical aperture** A_p is the actual area of the opening (dish, horn mouth). The ratio $\varepsilon_{ap} = A_e/A_p \leq 1$ is the **aperture efficiency**.

Aperture and directivity: for any antenna

$$D = \frac{4\pi}{\lambda^2} A_e = \frac{4\pi}{\lambda^2}\, \varepsilon_{ap} A_p, \qquad A_e\, \Omega_A = \lambda^2$$

where:

- λ = wavelength (m), Ω_A = beam area (sr).

How the aperture is set to increase the directivity:

1. **Make the aperture large in wavelengths.** D grows with A_p/λ^2. A bigger dish or horn, or the same dish at a higher frequency, gives a narrower beam (HPBW $\approx 51^\circ\, \lambda/a$ for a uniform opening of width a; about $70^\circ\, \lambda/d$ for a dish of diameter d).
2. **Illuminate it uniformly in amplitude.** A uniform field over the whole opening gives the largest A_e ($\varepsilon_{ap} \to 1$). A field tapered towards the edges wastes part of the area (but lowers the side lobes, a trade-off).
3. **Keep the phase uniform (in phase over the opening).** Then all parts add up along the axis; a phase error (a badly shaped dish, a feed off the focus) spreads the beam and lowers D.
4. **Avoid blockage and spill-over.** A feed or struts in front of the dish, and feed power that misses the dish, reduce ε_{ap}.

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Example

a 3 m dish at 10 GHz ($\lambda = 0.03$ m) with $\varepsilon_{ap} = 0.6$: $A_e = 0.6 \times 7.07 = 4.24$ m^2, so $D = 4\pi \times 4.24/0.03^2 = 5.92 \times 10^4$ (47.7 dBi). At 20 GHz the same dish gives 53.8 dBi.

3. Keyword Explanation

1. Aperture efficiency

The fraction of the physical area that is effective, $\varepsilon_{ap} = A_e/A_p$; typically 0.5–0.8 for a dish.

4. Notes and Remarks

- A thin half-wave dipole has almost no physical area but $A_e = 0.13\lambda^2$ (1(a)), so "aperture" means the effective area, not the metal.

1.(c) Describe how passive remote sensing is used to detect object. Marks: 5

1. Theory

1. Everything glows (Core Idea)

Any object warmer than absolute zero gives off electromagnetic noise, because its charges jiggle randomly. At microwave frequencies the power it sends is proportional to its temperature. A **radiometer** is a very sensitive receiver that measures this noise power. It sends nothing itself: that is why the method is **passive**.

2. Brightness temperature

A perfect absorber (a black body) at temperature T emits the most. A real surface emits only a fraction e (its **emissivity**, $0 \le e \le 1$) and reflects the rest of what falls on it. So it "looks" as warm as

$$T_B = e\,T_{phys} \quad \text{(plus the reflected sky)}$$

This is its **brightness temperature**. Soil has $e \approx 0.9$, so it looks warm (about 260 K). Calm water has $e \approx 0.4$ and looks cooler. Metal has $e \approx 0$: it emits almost nothing and mirrors the cold sky (a few kelvin), so it looks very cold.

2. Answer

Passive remote sensing detects and maps objects by measuring the natural thermal (noise) radiation they emit, with a radiometer, without sending any signal.

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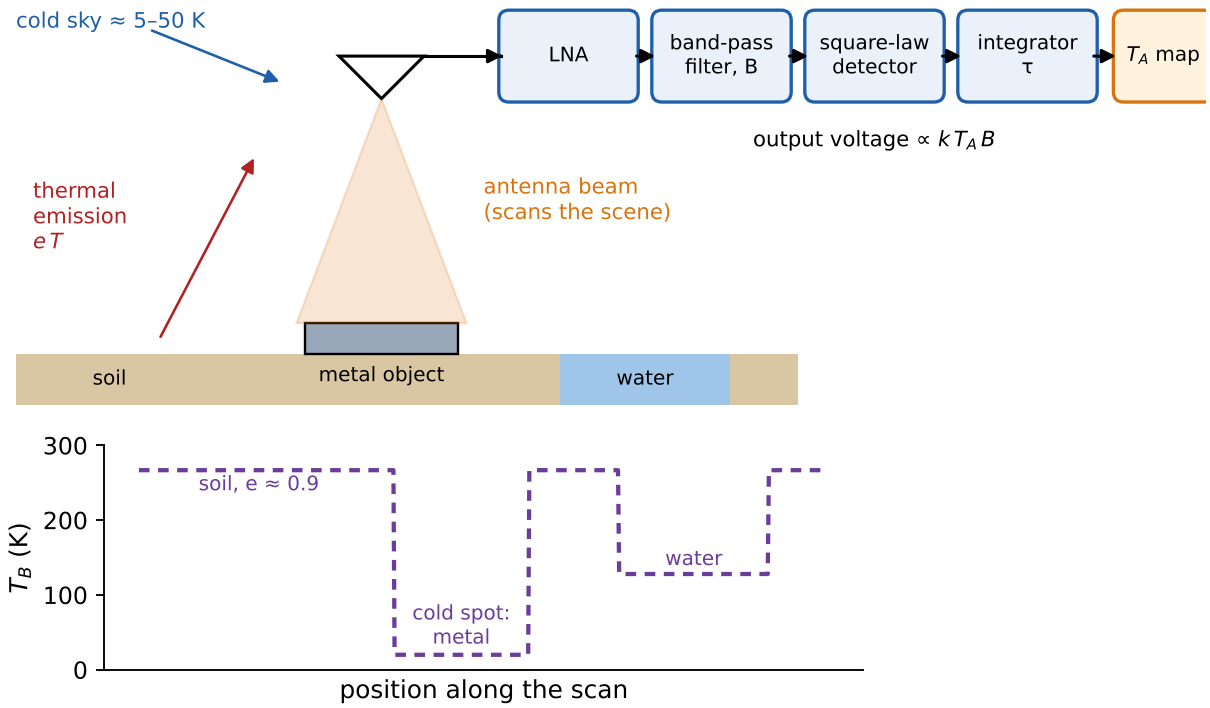


Figure 4: Passive remote sensing with a microwave radiometer

Reading the figure. The antenna beam (orange shading) looks down at soil, a metal object and water; the soil emits strongly (red arrow), the metal reflects the cold sky (blue arrow). The receiver chain amplifies, filters, detects and averages the noise. The plot below, in purple short dashes, is the brightness temperature along the scan: the metal object shows as a deep cold spot.

How an object is detected:

1. **Emission:** each part of the scene emits noise with brightness temperature $T_B = e\,T_{phys}$; objects with a different emissivity or temperature have a different T_B.
2. **Reception:** the antenna collects it. Its output is the **antenna temperature**, the T_B of the scene averaged over the beam:

$$T_A = \frac{1}{\Omega_A} \iint T_B(\theta,\phi)\, P_n(\theta,\phi)\, d\Omega$$

3. **Measurement:** the radiometer (LNA, band-pass filter of bandwidth B, square-law detector, integrator over a time τ) gives a voltage proportional to the received power $P = k\,T_A\,B$.
4. **Detection:** as the beam scans, an object appears as a change (contrast) ΔT_B against its background. A metal vehicle or ship looks cold; oil on water, ice, wet soil and fires each have their own T_B.
5. **Sensitivity:** the smallest change it can see is $\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$. With $T_{sys} = 500$ K, $B = 100$ MHz and $\tau = 10$ ms, $\Delta T_{min} = 0.5$ K.

Uses: soil moisture, sea ice and sea temperature from satellites; detecting ships, vehicles and concealed metal objects; weather (water vapour, rain). It works day and night and through cloud, and, sending nothing, it cannot be detected or jammed easily.

3. Keyword Explanation

1. Emissivity

How well a surface emits compared with a perfect black body at the same temperature; $e = 1 - |\Gamma|^2$, where Γ is its reflection coefficient.

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2. Antenna temperature

The temperature of a resistor that would deliver the same noise power to the receiver as the antenna does.

1.(d) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$. Marks: 5

1. Theory

1. What we are given (Core Idea)

The power pattern is $U(\theta, \phi) = \sin\theta \sin\phi$, and it exists only for $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq \pi$ (a half-space, $y \geq 0$); elsewhere it is zero. Its peak is $U_{max} = 1$ at $\theta = \phi = 90°$, the $+y$ direction.

2. The tools

- **Half-power beamwidth (HPBW):** the angle between the two directions where U falls to half its peak.
- **Beamwidth between first nulls (FNBW):** the angle between the first zeros on either side of the peak.
- **Radiated power:** add up U over all directions, $P_{rad} = \iint U \sin\theta \, d\theta \, d\phi$.
- **Directivity:** $D = 4\pi U_{max}/P_{rad}$ (1(a)).

2. Answer

Given: $U(\theta, \phi) = \sin\theta \sin\phi$ for $0 \leq \theta \leq \pi$, $0 \leq \phi \leq \pi$ (zero elsewhere); $U_{max} = 1$ at $\theta = 90°, \phi = 90°$. Take U in W/sr (the pattern is given without a constant, so $U_m = 1$).

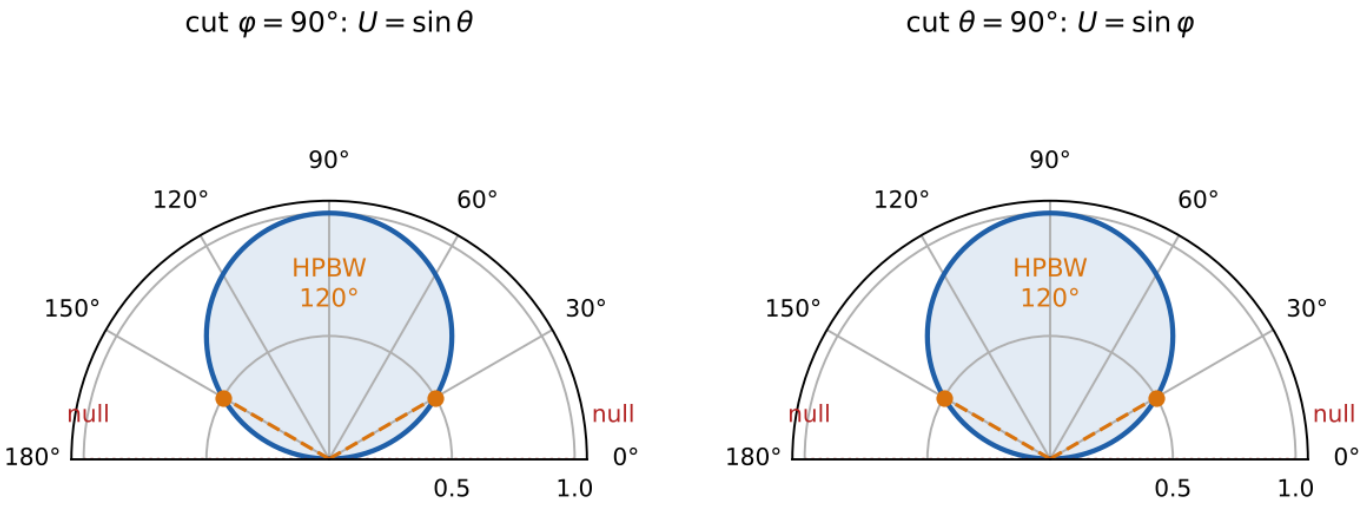


Figure 5: The pattern in its two principal planes

Reading the figure. Left: the cut $\phi = 90°$, where $U = \sin\theta$; right: the cut $\theta = 90°$, where $U = \sin\phi$. Both are the same circle-shaped lobe. The dashed orange lines go to the half-power points at 30° and 150°; the dotted red lines mark the nulls at 0° and 180°.

Step 1: Half-power beamwidth

In the plane $\phi = 90°$: $\sin\theta = \frac{1}{2}$ gives $\theta = 30°$ and 150°. The same in the plane $\theta = 90°$ for ϕ.

$$\text{HPBW} = 150° - 30° = 120° \quad \text{(in both principal planes)}$$

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$$\boxed{\theta_{HP} = \phi_{HP} = 120^\circ}$$

Step 2: Beamwidth between first nulls

$\sin\theta = 0$ at $\theta = 0^\circ$ and 180°; $\sin\phi = 0$ at $\phi = 0^\circ$ and 180°.

$$\boxed{\text{FNBW} = 180^\circ \ \text{(in both planes)}}$$

Step 3: Radiated power

$$P_{rad} = \int_0^\pi \int_0^\pi \sin\theta \, \sin\phi \, \sin\theta \, d\theta \, d\phi = \int_0^\pi \sin^2\theta \, d\theta \int_0^\pi \sin\phi \, d\phi = \frac{\pi}{2} \times 2 = \pi$$

$$\boxed{P_{rad} = \pi \text{ W} \approx 3.14 \text{ W} \ \ (\text{for } U_m = 1 \text{ W/sr})}$$

Step 4: Directivity

$$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi \times 1}{\pi} = 4$$

$$\boxed{D = 4 = 6.02 \text{ dBi}}$$

The beam area is $\Omega_A = P_{rad}/U_{max} = \pi$ sr, a quarter of the sphere.

Final Answer: HPBW $= 120^\circ$ and FNBW $= 180^\circ$ in both principal planes; $P_{rad} = \pi$ W (for $U_m = 1$); $D = 4$ (6.02 dBi).

3. Notes and Remarks

- **The approximate formula** $D \approx 41\,253/(\theta_{HP}\phi_{HP}) = 41\,253/120^2 = 2.86$ is well below the exact 4: the approximation is meant for narrow beams, and this beam is very wide.
- If the pattern is given as $U = U_m \sin\theta \sin\phi$, the radiated power is πU_m; the directivity is still 4.

4. Final Concise Answer

(a) Directivity $D = U_{max}/U_{av} = 4\pi U_{max}/P_{rad} = 4\pi/\Omega_A$ (isotropic 1, half-wave dipole 1.64). Effective height $h_e = V_{oc}/E = (1/I_0)\int I\,dz$, the length of an equivalent uniform-current antenna (λ/π for a half-wave dipole). Beam area $\Omega_A = \oint P_n \, d\Omega \approx \theta_{HP}\phi_{HP}$, the cone holding all the power at peak intensity.

(b) The effective aperture $A_e = P_r/S$ is the area that collects the received power; $D = 4\pi A_e/\lambda^2 = 4\pi\varepsilon_{ap}A_p/\lambda^2$. Directivity rises with an aperture that is large in wavelengths and uniformly illuminated in amplitude and phase, with little blockage and spill-over.

(c) A radiometer measures the natural thermal emission $T_B = eT$ of the scene (T_A averaged over the beam, $P = kT_AB$); an object shows as a contrast in T_B (metal looks cold), with a resolution $\Delta T = T_{sys}/\sqrt{B\tau}$; nothing is transmitted.

(d) HPBW $= 120^\circ$, FNBW $= 180^\circ$ in both planes; $P_{rad} = \pi$ W (for $U_m = 1$ W/sr); $D = 4\pi/\pi = 4$ (6.02 dBi).

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2. (a) Show that the normalized resistances represent circles in reflection coefficient plane. Marks: 6
   (b) Write the properties of r-circles and x-circles of a Smith chart. Marks: 6
   (c) Describe how TE mode is different from TM mode in a circular waveguide. Marks: 3
   (d) The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15+j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit [*sic*: should be short-circuit] is replaced by a load impedance $Z_L = 37 + j45$ Ω. Marks: 10

2.(a) Show that the normalized resistances represent circles in reflection coefficient plane. Marks: 6

1. Theory

1. Two ways to describe a load (Core Idea)

A load on a line of characteristic impedance Z_0 can be described by its **normalized impedance** $z = Z_L/Z_0 = r + jx$, or by its **reflection coefficient** $\Gamma = (z-1)/(z+1)$, the fraction of the wave it sends back. Both carry the same information. Every passive load has $r \geq 0$, which means $|\Gamma| \leq 1$: all of them fit inside a circle of radius 1 in the Γ plane.

2. What the question asks

Write $\Gamma = u + jv$. If we fix r (say $r = 1$) and let x take every value, the points Γ do not scatter: they lie on a circle. Drawing these circles for many values of r (and similar circles for x) gives the **Smith chart**. The same derivation was done in full in 2(a) of the 2020–21 paper.

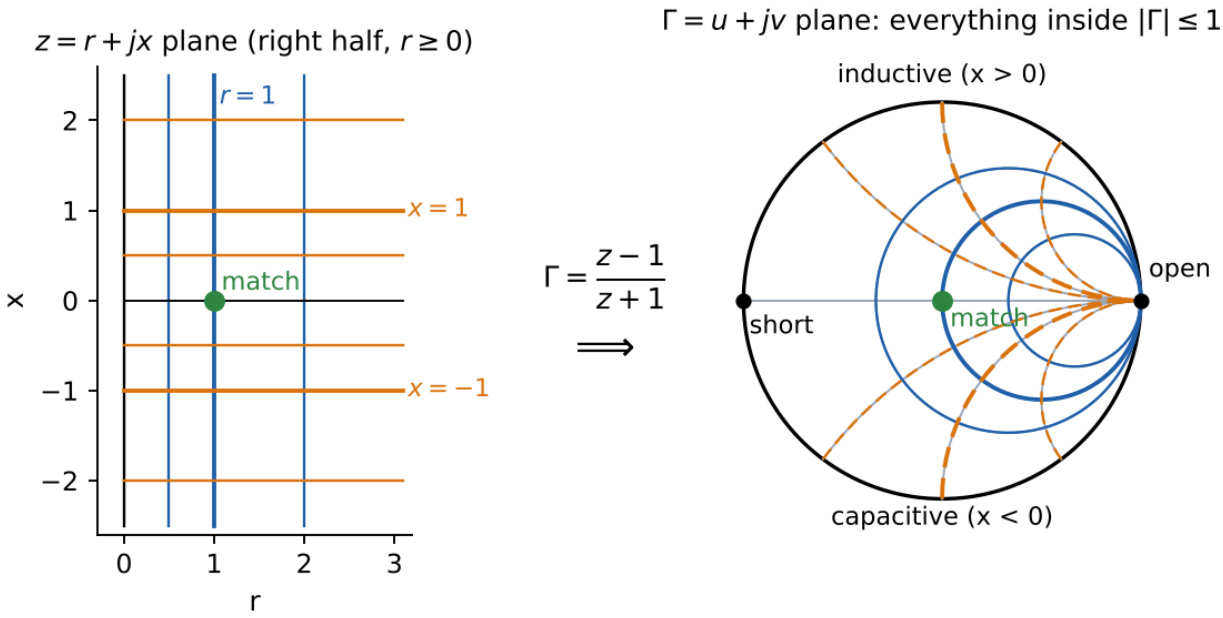


Figure 6: The z plane maps onto the inside of the unit circle of the Gamma plane

Reading the figure. Left: the right half of the z plane, with vertical lines of constant r (solid blue) and horizontal lines of constant x (solid orange). Right: the same lines in the Γ plane, where they become circles and arcs inside $|\Gamma| = 1$; the match $z = 1$ goes to the centre.

2. Answer

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Step 1: Invert the relation

Start from $\Gamma = \dfrac{z-1}{z+1}$, where $z = r + jx$ is the normalized impedance and $\Gamma = u + jv$. Cross-multiply and collect the z terms:

$$\Gamma z + \Gamma = z - 1 \;\Rightarrow\; z(1-\Gamma) = 1 + \Gamma \;\Rightarrow\; z = \frac{1+\Gamma}{1-\Gamma}$$

Put in $\Gamma = u + jv$:

$$r + jx = \frac{(1+u) + jv}{(1-u) - jv}$$

Step 2: Separate the real part

Multiply the top and bottom by the conjugate of the denominator, $(1-u) + jv$:

$$r + jx = \frac{[(1+u) + jv][(1-u) + jv]}{(1-u)^2 + v^2}$$

Multiply out the top, using $j^2 = -1$:

$$(1+u)(1-u) + jv(1+u) + jv(1-u) + j^2v^2 = (1 - u^2 - v^2) + j2v$$

so

$$r + jx = \frac{(1 - u^2 - v^2) + j2v}{(1-u)^2 + v^2}$$

The real parts of the two sides must be equal:

$$r = \frac{1 - u^2 - v^2}{(1-u)^2 + v^2}$$

Step 3: Rearrange

Cross-multiply and expand $(1-u)^2$:

$$r(1 - 2u + u^2 + v^2) = 1 - u^2 - v^2$$

Open the bracket and bring the u and v terms to the left:

$$r - 2ru + ru^2 + rv^2 + u^2 + v^2 = 1$$

$$(1+r)u^2 - 2ru + (1+r)v^2 = 1 - r$$

Divide by $(1+r)$:

$$u^2 - \frac{2r}{1+r}u + v^2 = \frac{1-r}{1+r}$$

Step 4: Complete the square

In u. Half of the coefficient of u is $r/(1+r)$, so add $r^2/(1+r)^2$ to both sides:

$$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \frac{1-r}{1+r} + \frac{r^2}{(1+r)^2}$$

Put the right side over the denominator $(1+r)^2$:

$$\frac{(1-r)(1+r) + r^2}{(1+r)^2} = \frac{1 - r^2 + r^2}{(1+r)^2} = \frac{1}{(1+r)^2}$$

Therefore

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$$\boxed{\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2}$$

This is the equation of a circle in the u–v (Γ) plane with

$$\text{centre } \left(\frac{r}{1+r},\ 0\right), \qquad \text{radius } \frac{1}{1+r}$$

So every normalized resistance r is a circle in the reflection coefficient plane. For example $r = 0$: centre $(0, 0)$, radius 1 (the unit circle); $r = 1$: centre $(0.5, 0)$, radius 0.5; $r \to \infty$: the point $(1, 0)$.

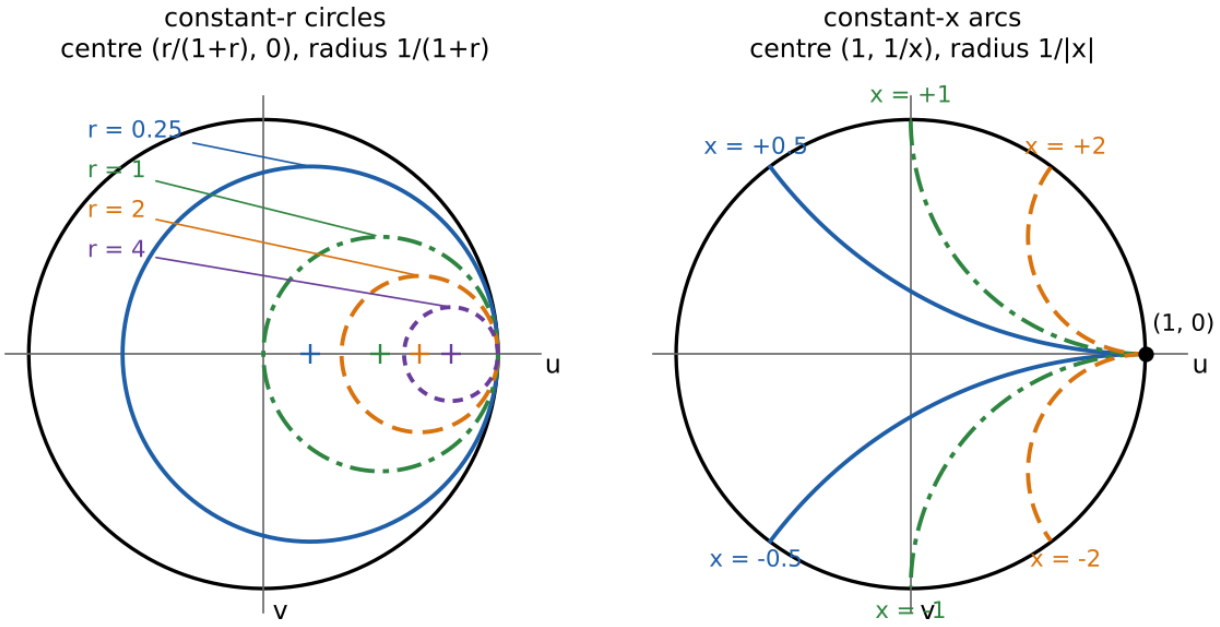


Figure 7: Constant-r circles and constant-x arcs

Reading the figure. Left: constant-r circles for $r = 0.25, 1, 2, 4$; their centres (+) lie on the u axis and every circle touches the point $(1, 0)$. Right: the constant-x arcs, used in 2(b).

Steps to remember

Invert the relation → Separate the real part → Rearrange → Complete the square.

2.(b) Write the properties of r-circles and x-circles of a Smith chart. Marks: 6

1. Theory

The basics

The Smith chart is a map of the reflection coefficient $\Gamma = u + jv$, whose grid is drawn in normalised impedance $z = r + jx = (1 + \Gamma)/(1 - \Gamma)$. So every impedance has one point on the chart. Holding r fixed and letting x change traces one curve, and holding x fixed traces another. 2(a) shows that these curves are circles.

1. The x-circles (Core Idea)

The imaginary part of Step 2 in 2(a), $x = \dfrac{2v}{(1-u)^2 + v^2}$, rearranges the same way into

$$(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$$

a circle with centre $(1, 1/x)$ and radius $1/|x|$. Only the part inside $|\Gamma| = 1$ is used, so on the chart it appears as an arc.

2. Why the properties matter

The Smith chart is these two families of circles drawn together. Their properties tell us where to find a short, an open, a match, an inductive or capacitive load, and how a point moves as we go along the line.

2. Answer

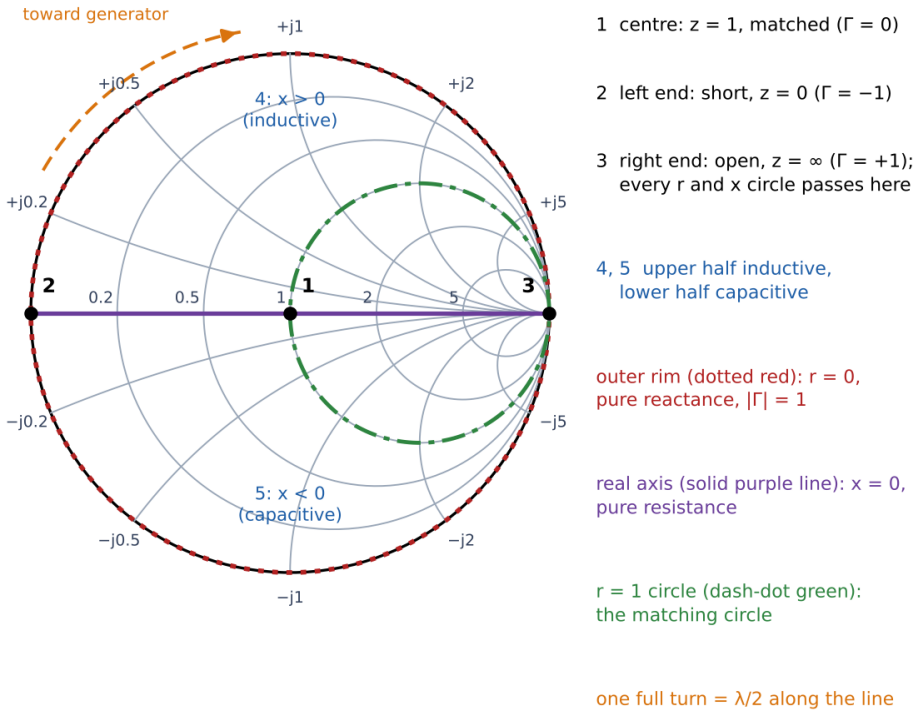


Figure 8: The landmarks of the Smith chart

Reading the figure. The numbered points and coloured lines are the properties listed below: the centre (1), the short (2) and the open (3); the upper (4) and lower (5) halves; the dotted red rim ($r = 0$), the solid purple real axis ($x = 0$) and the dash-dot green $r = 1$ circle. The orange arrow shows the direction "toward generator".

Properties of the r-circles (centre $\left(\frac{r}{1+r}, 0\right)$, radius $\frac{1}{1+r}$):

1. All centres lie on the real (u) axis.
2. Every r-circle passes through the point $(1, 0)$, and the circles touch each other there and nowhere else.
3. As r grows the circle shrinks: $r = 0$ is the outer unit circle (pure reactance, $|\Gamma| = 1$), $r = 1$ passes through the centre of the chart, $r \to \infty$ shrinks to the point $(1, 0)$ (open circuit).
4. All circles with $r \geq 0$ lie inside the unit circle: passive loads are inside the chart.

Properties of the x-circles (centre $\left(1, \frac{1}{x}\right)$, radius $\frac{1}{|x|}$):

1. All centres lie on the vertical line $u = 1$, outside the chart; only arcs inside the unit circle are drawn.
2. Every x-circle also passes through $(1, 0)$.
3. Positive x (inductive) arcs are in the upper half, negative x (capacitive) arcs in the lower half.
4. $x = 0$ is the real axis (a circle of infinite radius): pure resistance. As $|x|$ grows the arc shrinks towards $(1, 0)$.

Properties of the chart as a whole:

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- The centre $(r = 1, x = 0)$ is the matched load, $\Gamma = 0$; the left end $(-1, 0)$ is a short circuit, the right end $(1, 0)$ an open circuit.
- r-circles and x-circles cross at right angles.
- A circle about the centre is a circle of constant $|\Gamma|$ (constant SWR). Moving along a lossless line moves the point on this circle, clockwise towards the generator; one full turn is $\lambda/2$.
- Turning half a turn ($\lambda/4$) changes z to $1/z$, so the same chart reads admittances too.

2.(c) Describe how TE mode is different from TM mode in a circular waveguide. Marks: 3

1. Theory

1. Modes (Core Idea)

A hollow metal pipe can guide a wave only in certain field patterns, called **modes**. In a **TE** (transverse electric) mode the electric field has no component along the pipe ($E_z = 0$); in a **TM** (transverse magnetic) mode the magnetic field has none ($H_z = 0$). Each mode passes only above its own **cutoff frequency**. For a round pipe of radius a the patterns are described by Bessel functions, and each cutoff comes from a root of one of them.

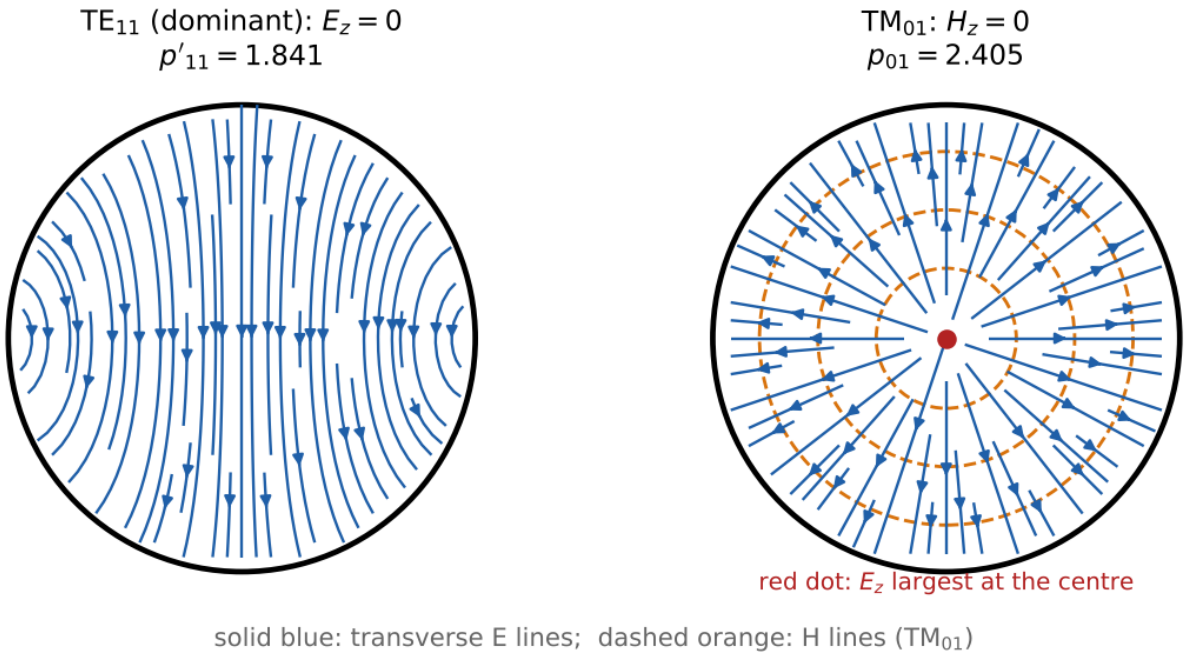


Figure 9: TE11 and TM01 in a circular waveguide

Reading the figure. Left: TE_{11}, whose electric lines (solid blue) run across the pipe from one side to the other, like the TE_{10} mode of a rectangular guide. Right: TM_{01}, whose electric lines are radial and whose magnetic lines (dashed orange) are circles; its E_z is largest on the axis (red dot).

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2. Answer

	TE_{nm} modes	TM_{nm} modes
Field along the guide	$E_z = 0,\ H_z \neq 0$	$H_z = 0,\ E_z \neq 0$
Cutoff condition	zero of $J_n'(k_c a)$: $k_c = p'_{nm}/a$	zero of $J_n(k_c a)$: $k_c = p_{nm}/a$
Lowest mode	TE_{11}, $p'_{11} = 1.841$: the **dominant** mode of the guide	TM_{01}, $p_{01} = 2.405$
Wave impedance	$Z_{TE} = k\eta/\beta$ (above η)	$Z_{TM} = \beta\eta/k$ (below η)
Special mode	TE_{01}: attenuation falls as frequency rises (long low-loss links)	TM_{01}: field symmetric about the axis (rotary joints)

Table 1: TE and TM modes in a circular waveguide

Both have the cutoff frequency $f_c = \dfrac{p\,c}{2\pi a\sqrt{\epsilon_r}}$, with $p = p'_{nm}$ for TE and $p = p_{nm}$ for TM. Since $p_{01} = 2.405 > p'_{11} = 1.841$, TE_{11} is the only mode between $f_{c,11}$ and $1.31 f_{c,11}$.

2.(d) The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit [*sic*: should be short-circuit] **is replaced by a load impedance $Z_L = 37 + j45$ Ω. Marks: 10**

1. Theory

Note on the paper

"sort-circuit" is a misprint for "short-circuit". The question is solved as intended.

1. A lossy line (Core Idea)

On a real line the wave both turns in phase and loses strength as it travels. Both effects are in the **propagation constant** $\gamma = \alpha + j\beta$: α (Np/m) is how fast the amplitude falls, β (rad/m) how fast the phase turns. The wave varies as $e^{-\gamma z} = e^{-\alpha z} e^{-j\beta z}$.

2. Input impedance with losses

For a lossless line the input impedance contains $\tan \beta l$. With losses the same formula holds with $j \tan \beta l$ replaced by $\tanh \gamma l$:

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l}$$

For a short circuit ($Z_L = 0$) only one term is left: $Z_{sc} = Z_0 \tanh \gamma l$. So a measured Z_{sc} gives $\tanh \gamma l$, and from it γ.

3. Undoing a tanh

If $\tanh \gamma l = T$, then $e^{2\gamma l} = \dfrac{1+T}{1-T}$, so

$$\gamma l = \frac{1}{2} \ln \frac{1+T}{1-T}$$

The log of a complex number $Ae^{j\psi}$ is $\ln A + j(\psi + 2n\pi)$: the real part gives αl, the imaginary part gives βl, but only up to whole multiples of π.

2. Answer

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Given:

- Length $l = 2$ m, characteristic impedance $Z_0 = 75\ \Omega$
- Short-circuit input impedance $Z_{sc} = 15 + j120\ \Omega$
- New load $Z_L = 37 + j45\ \Omega$

Relationships:

$$Z_{sc} = Z_0 \tanh \gamma l, \qquad Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l}$$

Step 1: Find tanh of gamma l

$$\tanh \gamma l = \frac{Z_{sc}}{Z_0} = \frac{15 + j120}{75} = 0.2 + j1.6$$

Step 2: Find gamma l

$$e^{2\gamma l} = \frac{1 + \tanh \gamma l}{1 - \tanh \gamma l} = \frac{1.2 + j1.6}{0.8 - j1.6} = -0.5 + j1.0 = 1.1180\angle 116.57°$$

$$2\gamma l = \ln 1.1180 + j(2.0344 + 2n\pi) \ \Rightarrow \ \gamma l = 0.05579 + j(1.0172 + n\pi)$$

Step 3: Alpha and beta

$$\alpha = \frac{0.05579}{2} = 0.02789 \text{ Np/m } (0.242 \text{ dB/m})$$

$$\beta = \frac{1.0172}{2} = 0.5086 \text{ rad/m} \quad (n = 0)$$

$$\boxed{\alpha = 0.0279 \text{ Np/m}, \qquad \beta = 0.509 \text{ rad/m}}$$

The smallest β ($n = 0$) is taken: the line is then $\beta l = 1.017$ rad $= 0.162\lambda$ long. A line longer than $\lambda/2$ would give $\beta = 0.509 + n\pi/2$ rad/m (2.079, 3.650, … rad/m); one measurement cannot tell them apart.

Step 4: Input impedance with the new load

$\tanh \gamma l = 0.2 + j1.6$ is already known, so:

$$Z_L + Z_0 \tanh \gamma l = (37 + j45) + 75(0.2 + j1.6) = 52 + j165$$

$$Z_0 + Z_L \tanh \gamma l = 75 + (37 + j45)(0.2 + j1.6) = 10.4 + j68.2$$

$$Z_{in} = 75 \times \frac{52 + j165}{10.4 + j68.2} = 185.85 - j28.84\ \Omega$$

$$\boxed{Z_{in} = 185.9 - j28.8\ \Omega = 188.1\angle -8.82°\ \Omega}$$

This result does not depend on the choice of n, since only $\tanh \gamma l$ enters.

Final Answer: $\alpha = 0.0279$ Np/m (0.242 dB/m), $\beta = 0.509$ rad/m; with $Z_L = 37 + j45\ \Omega$, $Z_{in} = 185.9 - j28.8\ \Omega$.

3. Notes and Remarks

- **Check:** substituting $\gamma l = 0.05579 + j1.0172$ back gives $\tanh \gamma l = 0.2 + j1.6$, and the computed Z_{in} is the same.
- With an open-circuit measurement as well, $Z_0 = \sqrt{Z_{oc} Z_{sc}}$ could be found too; here Z_0 was given.

4. Final Concise Answer

(a) From $z = \frac{1+\Gamma}{1-\Gamma}$ with $\Gamma = u + jv$: $r = \frac{1-u^2-v^2}{(1-u)^2+v^2}$, which rearranges to $\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2$: a circle with centre $(r/(1+r), 0)$ and radius $1/(1+r)$.

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(b) r-circles: centres on the real axis, all through (1, 0), shrinking from the unit circle ($r = 0$) to the point (1, 0) ($r = \infty$), $r = 1$ through the centre. x-circles: centres on $u = 1$, radius $1/|x|$, all through (1, 0), $x > 0$ above (inductive), $x < 0$ below (capacitive), $x = 0$ the real axis. They cross at right angles; centre = match, left = short, right = open; a constant-$|\Gamma|$ circle turns once per $\lambda/2$.

(c) TE: $E_z = 0$, cutoff from zeros of J_n', dominant TE_{11} ($p'_{11} = 1.841$), $Z_{TE} > \eta$. TM: $H_z = 0$, cutoff from zeros of J_n, lowest TM_{01} ($p_{01} = 2.405$), $Z_{TM} < \eta$.

(d) $\tanh \gamma l = 0.2 + j1.6$, $\gamma l = 0.0558 + j1.017$: $\alpha = 0.0279$ Np/m, $\beta = 0.509$ rad/m; $Z_{in} = 75\frac{52+j165}{10.4+j68.2} = 185.9 - j28.8\ \Omega$.

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3. (a) Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. Marks: 6

(b) Explain the condition for constant gain for amplifier. Marks: 5

(c) For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation. Marks: 5

(d) A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S. Marks: 9

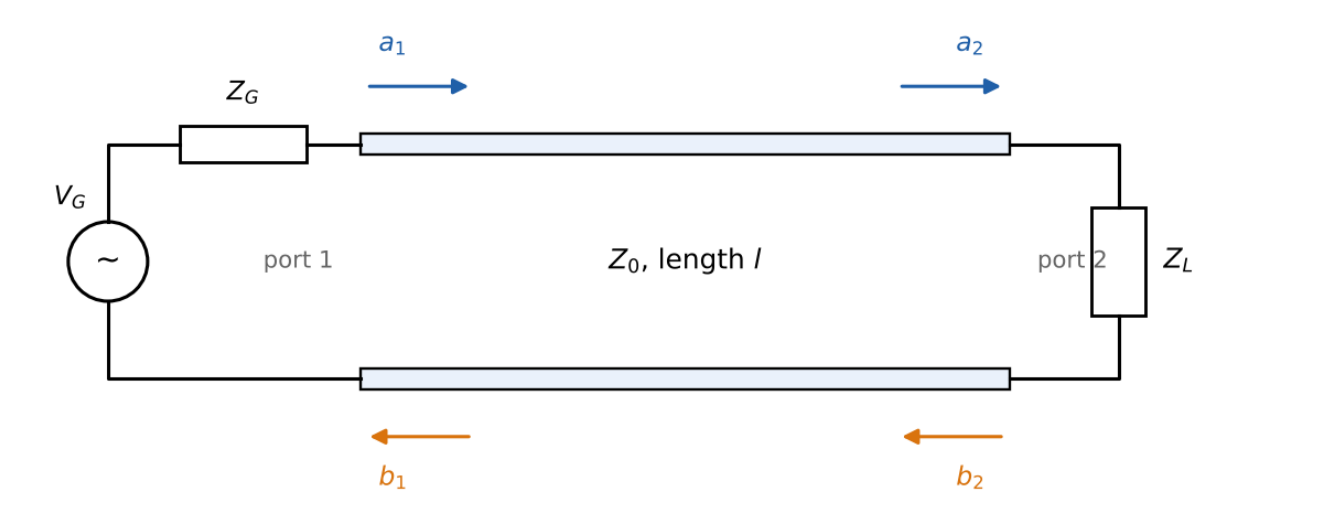


Figure 10: Figure 1 of the paper (redrawn)

3.(a) Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. Marks: 6

1. Theory

1. Why voltages and currents stop working (Core Idea)

At low frequency we describe a network with voltages and currents (Z or Y parameters), measured with ports open or shorted. At microwave frequencies three things go wrong: the voltage changes along every wire (the wires are lines), a true open or short is hard to make (an open radiates, a fringing capacitance appears), and a transistor terminated in an open or a short may oscillate. What *can* be measured easily is the power in a wave going into a port and the wave coming out of it, with every port ended in a matched load.

2. Waves in, waves out

At each port n we write the incident wave $a_n = V_n^+/\sqrt{Z_0}$ and the reflected (outgoing) wave $b_n = V_n^-/\sqrt{Z_0}$. Then $|a_n|^2/2$ is the power going in and $|b_n|^2/2$ the power coming out. The scattering parameters say how each incident wave turns into outgoing waves.

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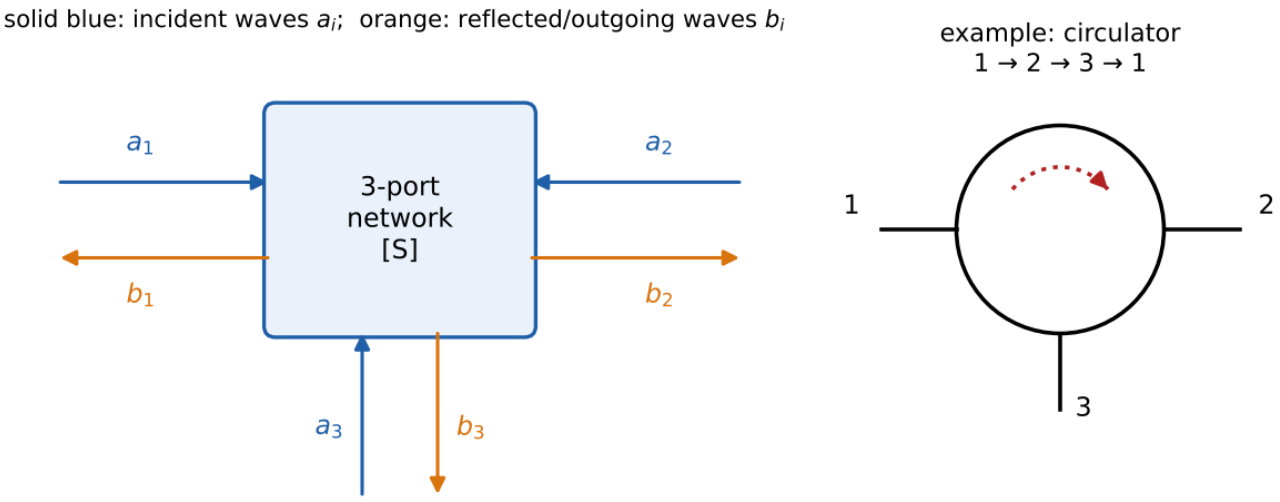


Figure 11: A three-port network with its incident and outgoing waves

Reading the figure. Left: the three-port network; the solid blue arrows are the incident waves a_1, a_2, a_3 and the solid orange arrows the outgoing waves b_1, b_2, b_3. Right: a circulator, the example below, which passes power around in the order $1 \to 2 \to 3 \to 1$.

2. Answer

Why S-parameters are used at microwave frequencies:

1. They are defined with matched loads (Z_0) at the ports, which are easy to make at microwave frequencies; open and short circuits are not.
2. Active devices stay stable with matched terminations; opens and shorts can make them oscillate.
3. They are measured directly as power-wave ratios with a vector network analyser.
4. They give gain, return loss and insertion loss at once ($|S_{21}|^2$ is the power gain, $|S_{11}|$ the input reflection), and cascade easily with flow graphs.

S-matrix of a 3-port network:

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \qquad S_{ij} = \left.\frac{b_i}{a_j}\right|_{a_k=0,\ k\neq j}$$

where:

- S_{ii} = reflection coefficient at port i with the other ports matched; S_{ij} $(i \neq j)$ = transmission coefficient from port j to port i.

Example: the ideal circulator

Power entering port 1 leaves at port 2, port 2 goes to port 3, port 3 to port 1; every port is matched:

$$[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

so $b_2 = a_1$, $b_3 = a_2$, $b_1 = a_3$. It is lossless ($[S]$ is unitary) but non-reciprocal ($S_{21} \neq S_{12}$), made with ferrite. A second example is the equal-split Wilkinson divider, $[S] = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, which is reciprocal and matched but lossy (its resistor absorbs the power that comes back).

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3. Notes and Remarks

- **A useful fact:** a 3-port that is lossless and reciprocal cannot be matched at all three ports. To get all three, give up one: losses (Wilkinson) or reciprocity (circulator).

3.(b) Explain the condition for constant gain for amplifier. Marks: 5

1. Theory

1. Where the gain comes from (Core Idea)

A transistor amplifier is a transistor between two **matching networks**. The transistor itself has a gain $|S_{21}|^2$, but how much of the source's power actually gets in depends on how well the input is matched. A matching network that presents a reflection coefficient Γ_S to the transistor can add gain (by matching better) or remove it (by mismatching on purpose).

2. The unilateral case

If the transistor sends almost nothing backwards ($S_{12} \approx 0$, **unilateral**), the input and output can be treated separately. The transducer gain splits into three factors:

$$G_{TU} = G_S\, G_0\, G_L, \qquad G_S = \frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2}, \quad G_0 = |S_{21}|^2, \quad G_L = \frac{1-|\Gamma_L|^2}{|1-S_{22}\Gamma_L|^2}$$

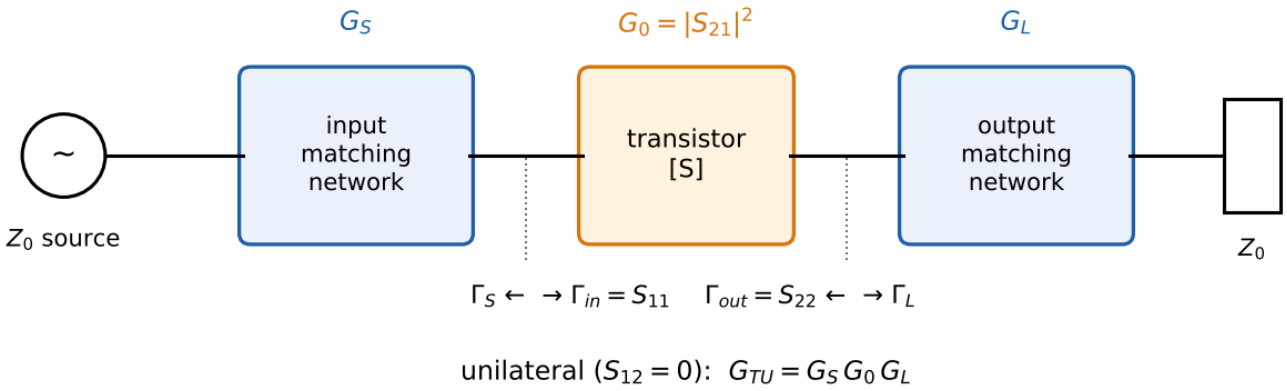


Figure 12: A unilateral amplifier: two matching networks around the transistor

Reading the figure. The source and load are Z_0. The input matching network sets Γ_S (gain block G_S), the transistor gives $G_0 = |S_{21}|^2$, the output network sets Γ_L (G_L). In the unilateral case the transistor's input reflection is just S_{11} and its output reflection S_{22}.

2. Answer

Maximum source gain. G_S is largest when the input is conjugately matched, $\Gamma_S = S_{11}^*$:

$$G_{Smax} = \frac{1}{1-|S_{11}|^2}$$

This is finite only if $|S_{11}| < 1$, which holds for an unconditionally stable transistor.

Constant gain. Every other Γ_S gives less. Define the normalized gain

$$g_s = \frac{G_S}{G_{Smax}} = \frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2}(1-|S_{11}|^2), \qquad 0 \le g_s \le 1$$

Condition: all source reflection coefficients Γ_S that give the same gain G_S lie on a circle in the Γ_S plane (the Smith chart). Fixing g_s and rearranging the equation above gives $|\Gamma_S - d_{gs}|^2 = r_{gs}^2$, with

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$$d_{gs} = \frac{g_s\, S_{11}^*}{1 - |S_{11}|^2(1 - g_s)}, \qquad r_{gs} = \frac{\sqrt{1 - g_s}\,(1 - |S_{11}|^2)}{1 - |S_{11}|^2(1 - g_s)}$$

where:

- d_{gs} = centre of the constant-gain circle, r_{gs} = its radius.

So a matching network gives the desired constant gain G_S when its Γ_S lies on that circle. The same holds at the output with S_{22}, Γ_L and g_L.

Key Points:

- $g_s = 1$ ($G_S = G_{Smax}$): the circle shrinks to the single point S_{11}^*.
- All centres lie on the line from the chart centre to S_{11}^*; a smaller gain gives a bigger circle closer to the centre.
- The circle $G_S = 1$ (0 dB) always passes through the chart centre ($\Gamma_S = 0$).
- **Conditions for using them:** the transistor is unilateral ($S_{12} \approx 0$; check with the unilateral figure of merit) and unconditionally stable ($|S_{11}| < 1$, $|S_{22}| < 1$, $k > 1$).

3.(c) For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation. Marks: 5

1. Theory

1. What a flow graph is (Core Idea)

A signal flow graph is a picture of the wave equations. Every wave (a_1, b_1, …) is a **node**. An arrow (a **branch**) from one node to another carries a factor: "the wave at the head equals the factor times the wave at the tail". A node equals the sum of all branches coming into it. With this picture, a network's response can be read off with a few simple rules instead of solving equations.

2. The pieces of Fig. 1

- **Source:** a generator V_G with impedance Z_G launches a wave b_s and reflects returning waves with $\Gamma_S = \dfrac{Z_G - Z_0}{Z_G + Z_0}$.
- **Line of length l:** a wave going through it is delayed in phase by $e^{-j\beta l}$, in either direction.
- **Load:** it reflects the wave arriving at it with $\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$.

2. Answer

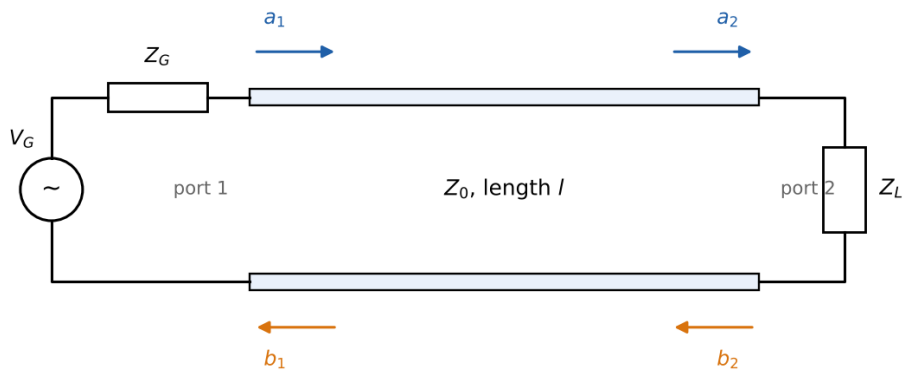


Figure 13: Repeated from the question: Figure 1 of the paper (redrawn)

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Step 1: One equation per branch

with the waves of Fig. 1 (a_1, a_2 travelling towards the load; b_2, b_1 travelling back):

$$a_1 = b_s + \Gamma_S b_1, \qquad a_2 = e^{-j\beta l} a_1, \qquad b_2 = \Gamma_L a_2, \qquad b_1 = e^{-j\beta l} b_2$$

Step 2: Draw the flow graph

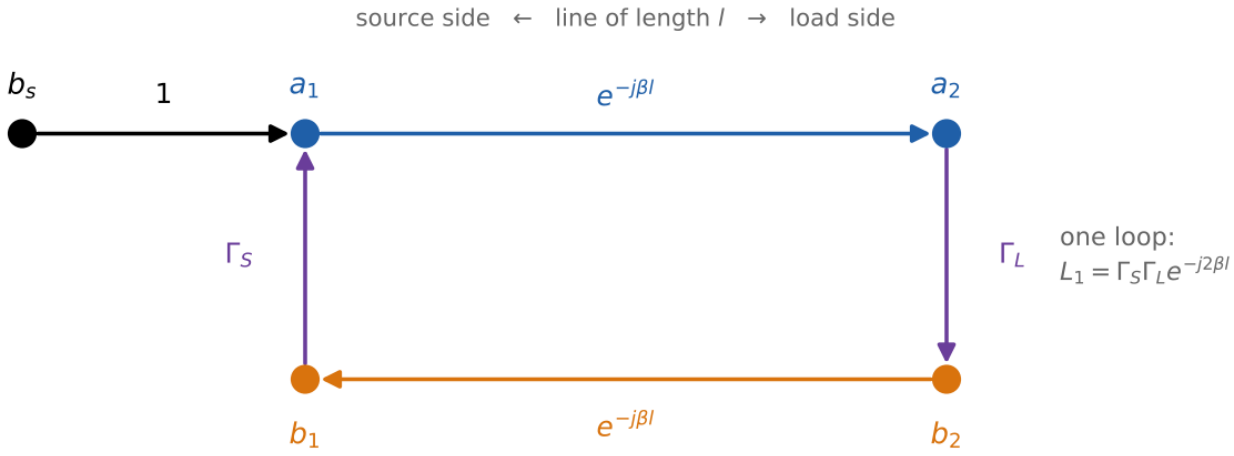


Figure 14: Signal flow graph of Fig. 1

Reading the figure. The top row carries the forward wave from a_1 to a_2 through the line ($e^{-j\beta l}$, solid blue); the load turns it back (Γ_L, purple) into b_2; the bottom row carries it back to b_1 (orange); the source mismatch Γ_S (purple) feeds it into a_1 again. The source wave b_s enters at a_1.

Step 3: Input reflection coefficient at port 1

Following the path from a_1 to b_1 (it does not use Γ_S, which is outside the port):

$$\Gamma_{in} = \frac{b_1}{a_1} = e^{-j\beta l} \cdot \Gamma_L \cdot e^{-j\beta l} = \Gamma_L\, e^{-j2\beta l}$$

Step 4: Input impedance

At port 1, $Z_{in} = Z_0 \dfrac{1+\Gamma_{in}}{1-\Gamma_{in}}$. With $\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$, multiply the top and bottom by $(Z_L + Z_0)e^{j\beta l}$ and use $e^{jx} + e^{-jx} = 2\cos x$, $e^{jx} - e^{-jx} = 2j\sin x$ (from $e^{\pm jx} = \cos x \pm j \sin x$):

$$Z_{in} = Z_0 \frac{(Z_L + Z_0)e^{j\beta l} + (Z_L - Z_0)e^{-j\beta l}}{(Z_L + Z_0)e^{j\beta l} - (Z_L - Z_0)e^{-j\beta l}} = Z_0 \frac{2Z_L \cos\beta l + j2Z_0 \sin\beta l}{2Z_0 \cos\beta l + j2Z_L \sin\beta l}$$

Divide the top and bottom by $2\cos\beta l$:

$$\boxed{Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}}$$

Steps to remember

One equation per branch → flow graph → Γ_{in} = path $a_1 \to b_1$: $\Gamma_L e^{-j2\beta l}$ → $Z_{in} = Z_0(1+\Gamma_{in})/(1-\Gamma_{in})$ → put in Γ_L, Euler, divide by $2\cos\beta l$.

Interpretation: the input impedance depends only on the line and the load, not on Z_G. The source mismatch only changes the size of the wave: by Mason's rule, with the single loop $L_1 = \Gamma_S\Gamma_L e^{-j2\beta l}$, $a_1 = \dfrac{b_s}{1 - \Gamma_S\Gamma_L e^{-j2\beta l}}$.

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3. Keyword Explanation

1. Mason's rule

The transfer between two nodes is $\sum_k P_k \Delta_k / \Delta$, where P_k are the forward paths, $\Delta = 1 - \sum(\text{loops}) + \sum(\text{pairs of non-touching loops}) - \ldots$, and Δ_k is Δ without the loops that touch path k.

3.(d) A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S. Marks: 9

1. Theory

1. What sets an amplifier's gain? (Core Idea)

A microwave amplifier is described by its scattering parameters. s_{11} says how much of a wave entering the input is reflected. If the source does not match the input, some power bounces back and never gets amplified. So the gain depends on the source's reflection coefficient Γ_S, which we choose with a matching network.

2. How much gain does the input side add?

Assuming a unilateral device (no signal goes backward, $s_{12} \approx 0$), the input side contributes

$$G_S = \frac{1 - |\Gamma_S|^2}{|1 - s_{11}\Gamma_S|^2}$$

It is largest when the source is the conjugate of the input, $\Gamma_S = s_{11}^*$, which gives $G_{Smax} = 1/(1 - |s_{11}|^2)$.

3. Why do we get circles?

We rarely want the maximum, since a lower gain may give a better bandwidth or noise. All the values of Γ_S that give one chosen G_S lie on a circle in the Γ plane. Define the normalised gain $g_s = G_S/G_{Smax}$. Then the circle has centre d_{gs} and radius r_{gs}:

$$d_{gs} = \frac{g_s s_{11}^*}{1 - |s_{11}|^2(1 - g_s)}, \qquad r_{gs} = \frac{\sqrt{1 - g_s}\,(1 - |s_{11}|^2)}{1 - |s_{11}|^2(1 - g_s)}$$

As G_S falls, the circle grows. The derivation is in 3(b). The steps here are: find G_{Smax}; choose a few gains below it; for each, find g_s, d_{gs} and r_{gs}; draw the circles on a Smith chart.

2. Answer

Given:

- $f = 4$ GHz, $s_{11} = 0.7\angle 125°$, unconditionally stable, unilateral design assumed.

Step 1: Maximum source gain

$$G_{Smax} = \frac{1}{1 - |s_{11}|^2} = \frac{1}{1 - 0.49} = \frac{1}{0.51} = 1.961$$

$$\boxed{G_{Smax} = 1.961 = 2.92 \text{ dB}, \quad \text{at } \Gamma_S = s_{11}^* = 0.7\angle -125°}$$

Step 2: Circles for several values of G_S

For each G_S: $g_s = G_S(1 - |s_{11}|^2) = 0.51\,G_S$, then $d_{gs} = \dfrac{g_s s_{11}^*}{1 - 0.49(1 - g_s)}$ (angle always $-125°$) and $r_{gs} = \dfrac{0.51\sqrt{1 - g_s}}{1 - 0.49(1 - g_s)}$.

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G_S (dB)	G_S	g_s	$\lvert d_{gs}\rvert$ (at $-125°$)	r_{gs}
2.5	1.778	0.907	0.665	0.163
2.0	1.585	0.808	0.624	0.246
1.0	1.259	0.642	0.545	0.370
0	1.000	0.510	0.470	0.470
-1.0	0.794	0.405	0.400	0.555

Table 2: Constant source-gain circles for s_{11} = 0.7\angle125°

Example (0 dB)

$$g_s = 0.51,\ d_{gs} = \frac{0.51 \times 0.7}{1 - 0.49 \times 0.49} = \frac{0.357}{0.7599} = 0.470\angle -125°,\ r_{gs} = \frac{0.51\sqrt{0.49}}{0.7599} = 0.470.$$

Step 3: Sketch

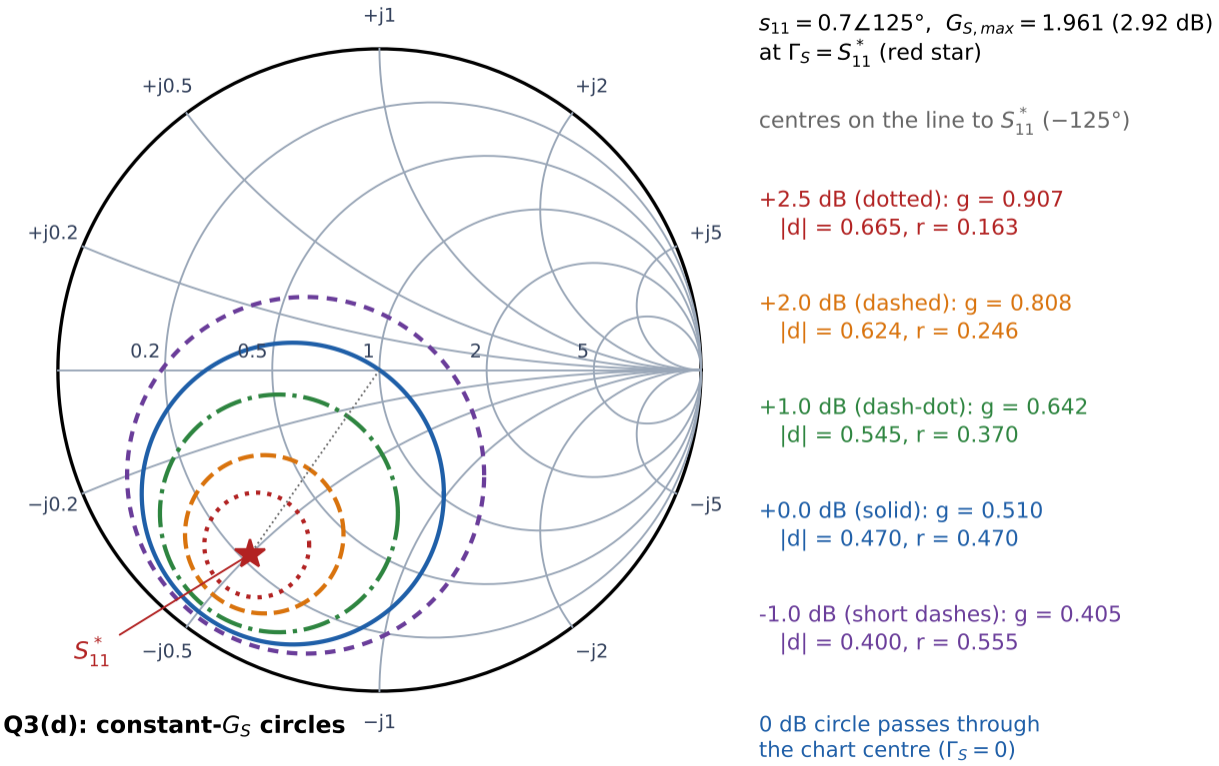


Figure 15: Constant source-gain circles on the Smith chart

Reading the figure. The star is s_{11}^*, where $G_S = G_{Smax} = 2.92$ dB. Around it, from small to large: +2.5 dB (dotted red), +2 dB (dashed orange), +1 dB (dash-dot green), 0 dB (solid blue) and −1 dB (purple short dashes). Their centres all lie on the grey line towards s_{11}^*, and the 0 dB circle passes through the chart centre.

Final Answer: $G_{Smax} = 1.961$ (2.92 dB) at $\Gamma_S = 0.7\angle -125°$; the circles of the table, centred on the line at $-125°$, grow and move towards the centre as G_S falls.

3. Notes and Remarks

- Any Γ_S on the 0 dB circle gives no gain or loss from the input network, the same as no matching at all ($\Gamma_S = 0$).
- The paper's formulas d_{gi} and r_{gi} are these with $i = s$ ($S_{ii} = S_{11}$).

4. Final Concise Answer

(a) At microwave frequencies opens and shorts are hard to make and destabilize devices, while matched loads and wave power are easy to measure (VNA), so networks are described by $b = [S]a$, $S_{ij} = b_i/a_j$ with the other ports matched. A 3-port: 3×3 matrix; example: the ideal circulator $[S] = [[0,0,1],[1,0,0],[0,1,0]]$ (matched, lossless, non-reciprocal).

(b) With $S_{12} \approx 0$, $G_{TU} = G_S G_0 G_L$; G_S is constant when Γ_S lies on the circle with centre $d_{gs} = g_s S_{11}^*/[1-|S_{11}|^2(1-g_s)]$ and radius $r_{gs} = \sqrt{1-g_s}(1-|S_{11}|^2)/[1-|S_{11}|^2(1-g_s)]$, $g_s = G_S(1-|S_{11}|^2) \le 1$; it needs an unconditionally stable, unilateral device; $g_s = 1$ is the point S_{11}^*.

(c) Flow graph: $b_s \to a_1 \xrightarrow{e^{-j\beta l}} a_2 \xrightarrow{\Gamma_L} b_2 \xrightarrow{e^{-j\beta l}} b_1 \xrightarrow{\Gamma_S} a_1$; $\Gamma_{in} = \Gamma_L e^{-j2\beta l}$, so $Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$.

(d) $G_{Smax} = 1/(1-0.49) = 1.961$ (2.92 dB) at $\Gamma_S = 0.7\angle -125°$; circles: 2 dB ($|d| = 0.624$, $r = 0.246$), 1 dB (0.545, 0.370), 0 dB (0.470, 0.470), -1 dB (0.400, 0.555), all centred at $-125°$.

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4. (a) Write short notes on the following topic

   i. Reflection coefficient
   ii. Standing wave ratio
   iii. RF radiation hazards and safety standard
   iv. Microwave cavity resonators Marks: 8

   (b) Show how quarter wavelength line is used for matching. Marks: 4

   (c) Explain how the RADAR range equation is used to detect object. Marks: 7

   (d) A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line. Marks: 6

4.(a) Write short notes on the following topic i. Reflection coefficient ii. Standing wave ratio iii. RF radiation hazards and safety standard iv. Microwave cavity resonators Marks: 8

1. Theory

1. What is a transmission line? (Core Idea)

At high frequency, a pair of wires is not a simple connection, because the wave takes time to travel along it. A voltage wave travels with a matching current wave. Their ratio is the line's **characteristic impedance** Z_0, for example 50 Ω.

2. Why is a wave reflected?

At the load the current and voltage must obey the load's own ratio Z_L. If $Z_L = Z_0$, the wave is absorbed. If not, the load cannot take the wave as it is, so part of it bounces back. That is the reflected wave, and Γ measures its size and phase.

3. What is a standing wave?

The forward and reflected waves add along the line. At some points they are in step and the voltage is large, and at others they oppose and it is small. This fixed pattern is a standing wave, and the ratio of its largest to smallest voltage is the SWR. With no reflection the pattern is flat.

4. Why are RF fields a hazard?

Body tissue holds water, and water absorbs microwave energy and warms up. The danger depends on the frequency and the power density, so standards set limits in those terms.

5. What is a cavity resonator?

Close a metal box on all sides, and a wave inside bounces between the walls. Only wavelengths that fit the box survive, just as only certain notes fit an organ pipe. So the box acts as a very sharp tuned circuit, with low loss.

2. Answer

i. Reflection coefficient

Definition: The reflection coefficient Γ at a point on a line is the ratio of the reflected voltage wave to the incident voltage wave at that point. At the load:

$$\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Key Points:

- It is a complex number with $0 \le |\Gamma| \le 1$ for a passive load: matched load $\Gamma = 0$, short $\Gamma = -1$, open $\Gamma = +1$.
- At a distance d from the load (lossless line): $\Gamma(d) = \Gamma_L e^{-j2\beta d}$; only its angle changes.
- $|\Gamma|^2$ is the fraction of the incident power reflected; the **return loss** is $RL = -20\log_{10}|\Gamma|$ dB.

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ii. Standing wave ratio

Definition: The incident and reflected waves add to a standing-wave pattern of $|V|$ along the line. The standing wave ratio is the ratio of its largest to its smallest value:

$$S = \frac{|V|_{max}}{|V|_{min}} = \frac{1+|\Gamma|}{1-|\Gamma|}, \qquad |\Gamma| = \frac{S-1}{S+1}$$

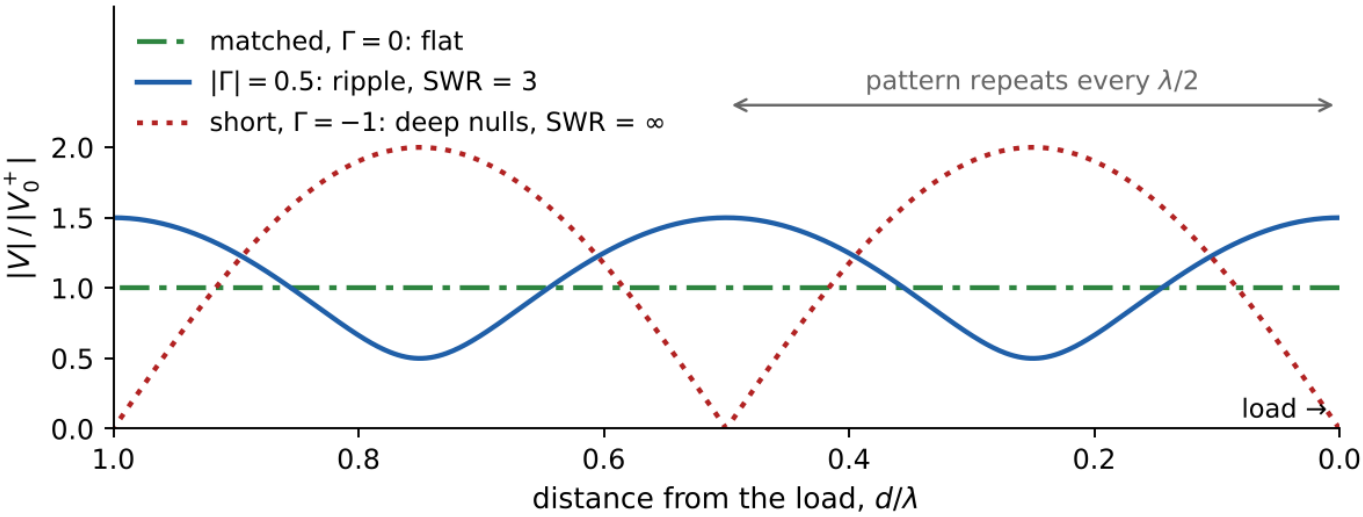


Figure 16: Standing-wave patterns for three loads

Reading the figure. The load is at the right. A matched load (dash-dot green) gives a flat $|V|$, SWR $= 1$; $|\Gamma| = 0.5$ (solid blue) ripples between 0.5 and 1.5, SWR $= 3$; a short (dotted red) swings between 0 and 2, SWR $= \infty$. Each pattern repeats every $\lambda/2$.

Key Points: $1 \le S < \infty$; successive maxima (or minima) are $\lambda/2$ apart and a maximum is $\lambda/4$ from a minimum; at a voltage minimum the impedance is real, Z_0/S, and at a maximum $Z_0 S$. SWR is easy to measure with a slotted line, which is how Z_L is found in 6(d).

iii. RF radiation hazards and safety standard

Hazards: RF energy absorbed by the body heats tissue (the main, **thermal** effect). Organs with little blood flow to carry the heat away, the lens of the eye (cataract) and the testes, are the most sensitive. Below about 10 MHz, induced currents can also stimulate nerves and cause shocks and burns. RF fields can also ignite fuel vapour, set off electro-explosive devices, and disturb pacemakers and other medical implants.

Measure of exposure: the **specific absorption rate**, SAR $= \sigma_t |E|^2/\rho$ (W/kg), the power absorbed per kilogram of tissue (σ_t = tissue conductivity in S/m, ρ = tissue density in kg/m^3, E = r.m.s. field in the tissue). Above about 6–10 GHz, where the energy is absorbed near the skin, the incident power density (W/m^2) is used.

Safety standards (ICNIRP, IEEE C95.1): limits averaged over 6 minutes, with a factor 5 between workers and the general public:

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Quantity	General public	Occupational
Whole-body average SAR	0.08 W/kg	0.4 W/kg
Local SAR, head and trunk (10 g)	2 W/kg	10 W/kg
Power density, 2–300 GHz (ICNIRP 1998)	10 W/m^2	50 W/m^2

Table 3: Basic restrictions and reference levels

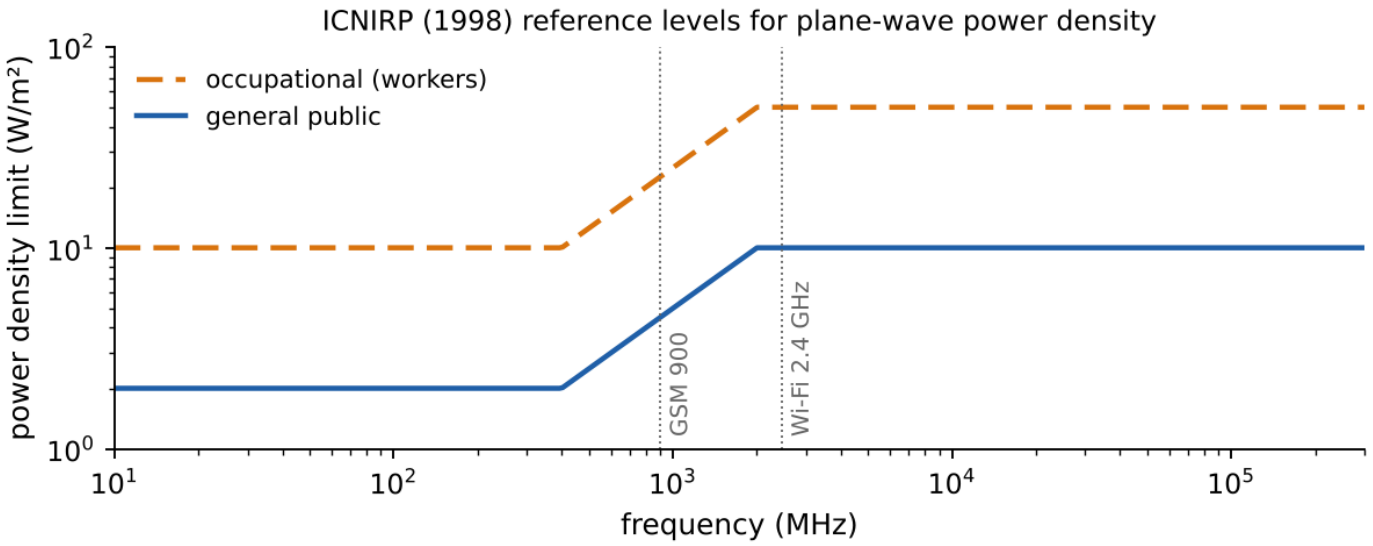


Figure 17: ICNIRP reference levels for power density

Reading the figure. The solid blue line is the general-public limit, the dashed orange line the occupational limit: flat at 2 and 10 W/m^2 from 10 to 400 MHz, rising as $f/200$ and $f/40$ up to 2 GHz, then flat at 10 and 50 W/m^2.

Protection: keep a safe distance (power density falls as $1/r^2$: an EIRP of 1 kW meets 10 W/m^2 beyond $r = \sqrt{1000/(4\pi \times 10)} = 2.8$ m), limit the time, use shielding and interlocks, and fence and sign the areas near transmitters.

iv. Microwave cavity resonators

Definition: A cavity resonator is a closed metal box (rectangular or cylindrical), or a length of waveguide shorted at both ends, in which the fields bounce back and forth and build up a standing wave at certain frequencies only. It is the microwave version of an LC tank circuit.

Resonant frequency of a rectangular cavity $a \times b \times d$ filled with a dielectric ϵ_r:

$$f_{mnp} = \frac{c}{2\sqrt{\mu_r \epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

The length d holds a whole number p of half guide-wavelengths. For $b < a < d$ the lowest (dominant) mode is TE$_{101}$.

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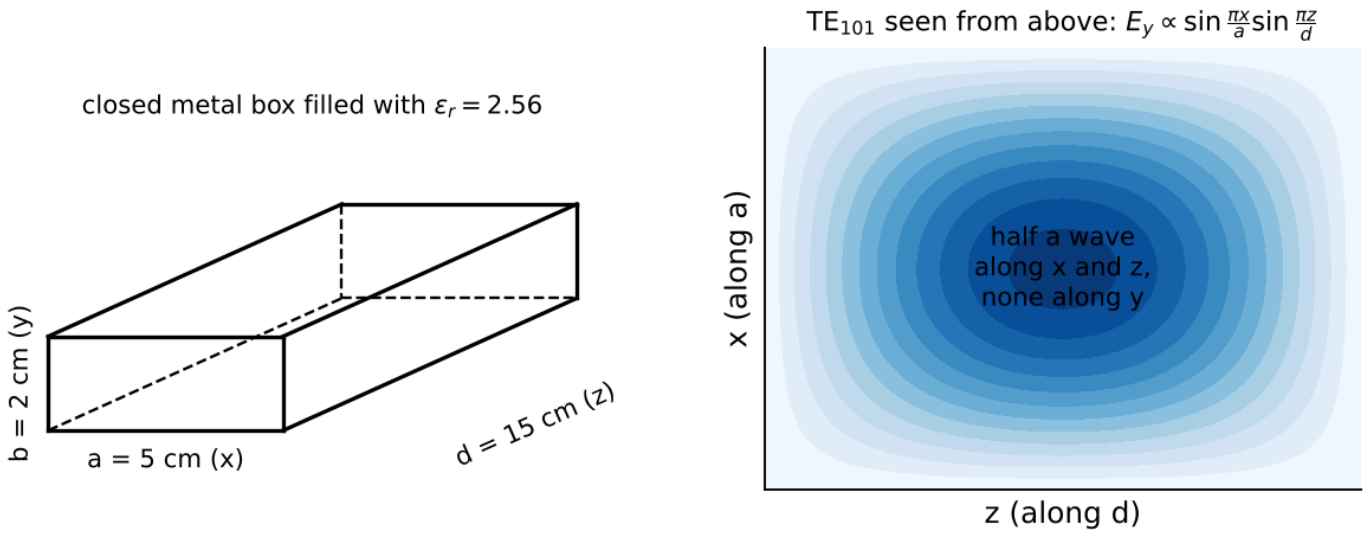


Figure 18: A rectangular cavity and its TE101 field

Reading the figure. Left: a closed rectangular box (the one solved in 6(c) of the 2020–21 paper). Right: the TE_{101} electric field seen from above, strongest in the middle and zero at the side walls: half a wave across a and half a wave along d.

Key Points: the quality factor $Q = \omega \dfrac{\text{energy stored}}{\text{power lost}}$ is very high (thousands to tens of thousands), so cavities are very selective. Uses: band-pass filters, oscillator tanks (klystrons, magnetrons), wavemeters to measure frequency, and frequency standards. They are tuned with a screw or a plunger that changes the size.

4.(b) Show how quarter wavelength line is used for matching. Marks: 4

1. Theory

1. Why matching matters (Core Idea)

When a line of impedance Z_0 feeds a load $R_L eZ_0$, part of the wave is reflected (4(a)): less power reaches the load and a standing wave appears on the line. We want something between the line and the load that makes the load "look like" Z_0 from the line, without wasting power. A short piece of a *different* line can do this.

2. A line changes the impedance it shows

A load Z_L seen through a lossless line of impedance Z_1 and length l looks like (derived in 3(c))

$$Z_{in} = Z_1 \frac{Z_L + jZ_1 \tan \beta l}{Z_1 + jZ_L \tan \beta l}$$

So by choosing the length and the impedance of the piece, we choose what the line sees.

3. The special length lambda/4

For $l = \lambda/4$, $\beta l = \pi/2$ and $\tan \beta l$ becomes infinite. Dividing the top and bottom by $\tan \beta l$ leaves only the j terms, and the formula becomes $Z_{in} = Z_1^2/Z_L$: the quarter-wave line turns an impedance "upside down" (a small Z_L looks large, a large one small). Another way to see it: the wave reflected at the far end of the section travels $\lambda/4$ there and $\lambda/4$ back, half a wavelength in all, so it returns turned by 180° and cancels the reflection from the near end, when Z_1 has the right value.

2. Answer

A section of line of characteristic impedance Z_1 and length $l = \lambda/4$ is placed between the main line (Z_0) and a resistive load R_L. Its input impedance is

$$Z_{in} = Z_1 \frac{R_L + jZ_1 \tan \beta l}{Z_1 + jR_L \tan \beta l}$$

With $\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$, $\tan \beta l \to \infty$, so

$$Z_{in} = \frac{Z_1^2}{R_L}$$

For a match we need $Z_{in} = Z_0$:

$$\boxed{Z_1 = \sqrt{Z_0 R_L}}$$

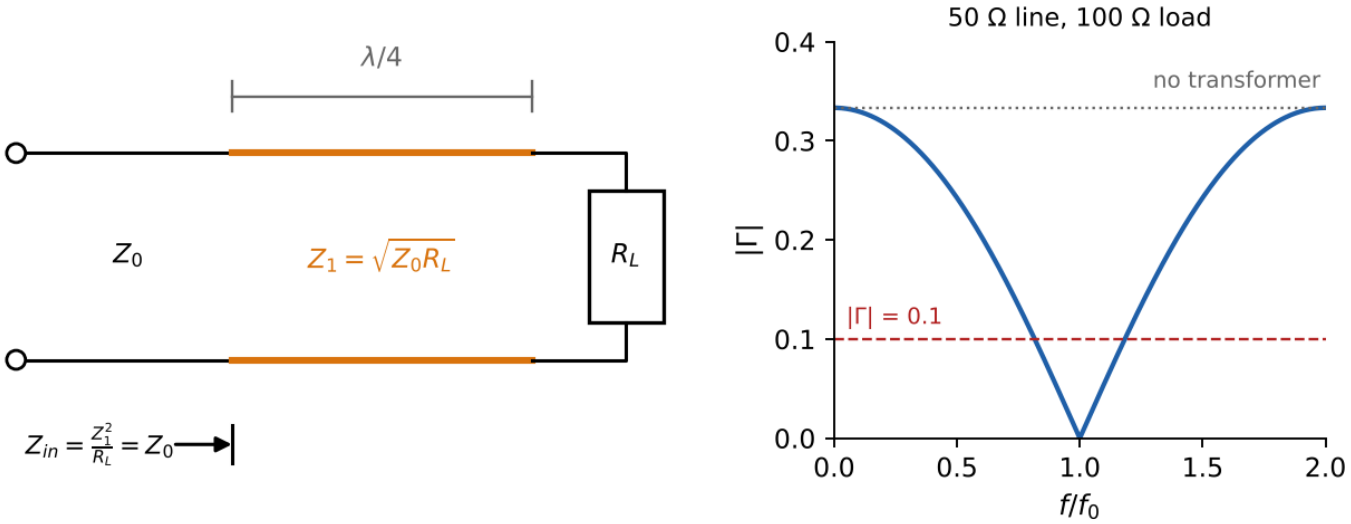


Figure 19: Quarter-wave transformer and its bandwidth

Reading the figure. Left: the thick orange $\lambda/4$ section of impedance $\sqrt{Z_0 R_L}$ between the line and R_L. Right: the reflection seen from the line for a 50 Ω line and a 100 Ω load (solid blue): zero at the design frequency f_0, rising on both sides.

Key Points: it works exactly only at one frequency (narrow band); the load must be real. A complex load is first moved along the line to a voltage maximum or minimum, where the impedance is real ($Z_0 S$ or Z_0/S), and the transformer is put there. Example: a 100 Ω load on a 50 Ω line needs $Z_1 = \sqrt{50 \times 100} = 70.7\ \Omega$.

4.(c) Explain how the RADAR range equation is used to detect object. Marks: 7

1. Theory

1. How radar sees (Core Idea)

A radar sends out a short, powerful pulse. When it hits an object, a tiny part bounces back. The radar listens for this echo: the time it takes gives the distance ($R = ct/2$), and the direction of the antenna gives the bearing. The **radar range equation** tells how strong the echo is, and so how far away an object can still be detected.

2. The spreading twice

The pulse spreads out over a sphere on its way to the target (power density $\propto 1/R^2$), and the echo spreads out again on the way back (another $1/R^2$). That is why the echo falls as $1/R^4$, very fast.

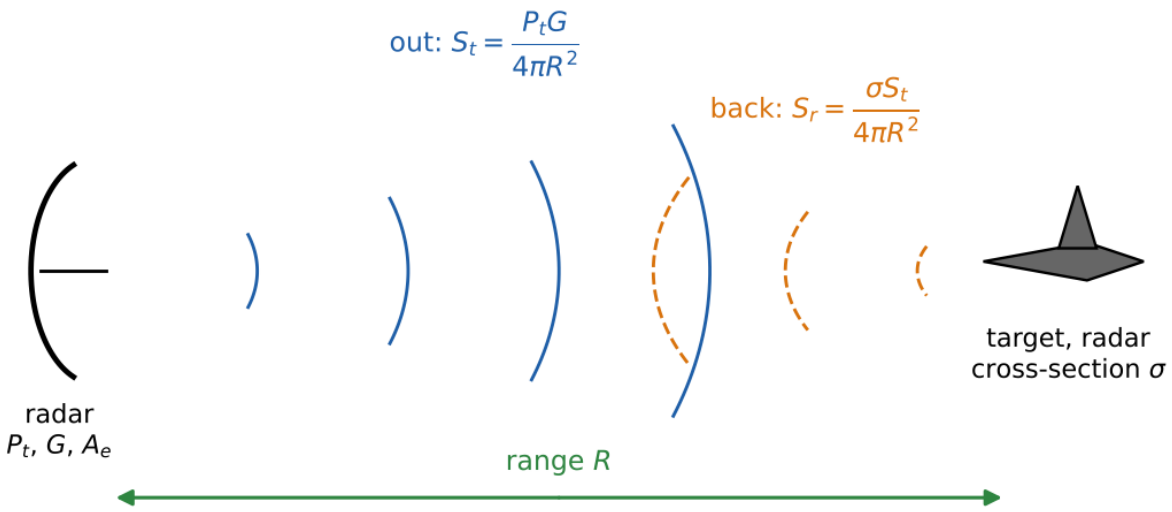


Figure 20: Radar geometry: the outgoing wave and the echo

Reading the figure. The radar on the left sends the outgoing wave (solid blue arcs) towards the target at range R. The target, with radar cross-section σ, sends back an echo (dashed orange arcs) that spreads again on its way back.

2. Answer

Derivation:

1. **Power density at the target:** a transmitter of power P_t with an antenna of gain G gives, at range R,
$$S_t = \frac{P_t G}{4\pi R^2} \ \text{W/m}^2$$
2. **Power scattered by the target:** the target intercepts and re-radiates (isotropically, by definition) the power σS_t, where σ is its **radar cross-section** (m^2).
3. **Power density back at the radar:**
$$S_r = \frac{\sigma S_t}{4\pi R^2} = \frac{P_t G \sigma}{(4\pi)^2 R^4}$$
4. **Received power:** the antenna with effective aperture $A_e = G\lambda^2/4\pi$ collects
$$P_r = S_r A_e = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

Use for detection: an object is detected when the echo is at least the smallest power the receiver can detect, $P_r \geq P_{min}$ (set by the receiver noise, $P_{min} = kT_s B \cdot (S/N)_{min}$). Putting $P_r = P_{min}$ gives the **maximum range**:

$$\boxed{R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}}$$

where:

- P_t = peak transmitted power (W), G = antenna gain, λ = wavelength (m), σ = radar cross-section (m^2), P_{min} = minimum detectable signal (W).

How it is used:

- **Range of the object:** from the echo delay, $R = ct/2$; the equation says whether an echo from that range is strong enough.

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- **Design:** it gives the transmitter power, antenna size and receiver sensitivity needed to see a target of a given σ at a given range.
- **Interpretation:** $R_{max} \propto P_t^{1/4}$, so doubling the range needs 16 times the power, or a bigger antenna (G^2), a more sensitive receiver, or pulse integration.
- **Target size:** a large or metallic target (large σ) is seen farther; stealth shapes reduce σ.

3. Keyword Explanation

1. Radar cross-section

The area that, intercepting the incident power density and radiating it equally in all directions, would give the same echo as the real target. It depends on the target's size, shape, material and the frequency.

4.(d) A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line. Marks: 6

1. Theory

1. The idea of a stub (Core Idea)

A mismatched load sends part of the wave back. As we move from the load towards the generator, the admittance seen on the line changes. At some point its real part becomes exactly Y_0: then $y = 1 + jb$. If we connect, at that point and in parallel, a short-circuited piece of line (a **stub**) whose input admittance is $-jb$, the two add to $y = 1$: a perfect match from there to the generator. The stub is chosen short-circuited because a short is easy to make and does not radiate. The same method is used in 2(b) of the 2020–21 paper.

2. Two numbers to find

- the **position** d: distance from the load to the stub, where $\mathrm{Re}(y) = 1$;
- the **length** l: the stub length that gives $-jb$. A short-circuited line of length l has $Z_{sc} = jZ_0 \tan \beta l$ (6(c)), so its normalized admittance is $y_{stub} = \dfrac{Z_0}{jZ_0 \tan \beta l} = -j \cot \beta l$: any pure susceptance can be made by choosing l.

3. Finding the position

At a distance d from the load, with $t = \tan \beta d$, the line shows $Z = Z_0 \dfrac{Z_L + jZ_0 t}{Z_0 + jZ_L t}$. Writing $Z_L = R_L + jX_L$ and taking $Y = 1/Z = G + jB$:

$$Y = \frac{(Z_0 - X_L t) + jR_L t}{Z_0 \left[R_L + j(X_L + Z_0 t)\right]} \;\Rightarrow\; G = \frac{R_L(1+t^2)}{R_L^2 + (X_L + Z_0 t)^2}, \quad B = \frac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0 \left[R_L^2 + (X_L + Z_0 t)^2\right]}$$

(multiply the top and bottom by the conjugate of the bracket and keep the real and imaginary parts). The match needs $G = Y_0 = 1/Z_0$:

$$Z_0 R_L(1+t^2) = R_L^2 + (X_L + Z_0 t)^2 \;\Rightarrow\; Z_0(R_L - Z_0)\, t^2 - 2X_L Z_0\, t + (Z_0 R_L - R_L^2 - X_L^2) = 0$$

a quadratic in t, whose two roots give the two possible stub positions:

$$t = \frac{X_L \pm \sqrt{R_L\left[(Z_0 - R_L)^2 + X_L^2\right]/Z_0}}{R_L - Z_0}$$

The same steps can be done graphically on the Smith chart (below).

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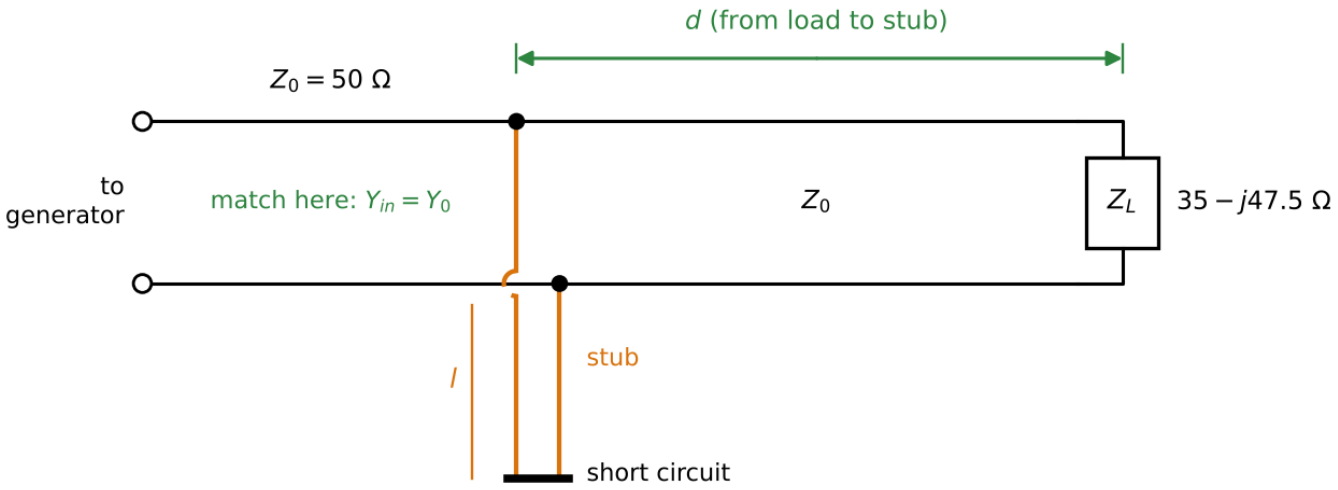


Figure 21: The stub between the load and the generator

Reading the figure. The load $35 - j47.5\ \Omega$ is at the right. At a distance d from it (solid green), the short-circuited stub of length l (orange) hangs in parallel. From the stub to the generator the line is matched.

2. Answer

Given:

- $Z_0 = 50\ \Omega$, $Z_L = 35 - j47.5\ \Omega$, short-circuited shunt stub of the same Z_0.

Step 1: Normalize

$$z_L = \frac{35 - j47.5}{50} = 0.70 - j0.95, \qquad y_L = \frac{1}{z_L} = 0.503 + j0.682$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.512\angle -78.33°, \qquad S = \frac{1 + 0.512}{1 - 0.512} = 3.09$$

Step 2: Position of the stub

With $t = \tan \beta d$, the condition $G = Y_0$ (Theory, part 3) gives

$$t = \frac{X_L \pm \sqrt{R_L\left[(Z_0 - R_L)^2 + X_L^2\right]/Z_0}}{R_L - Z_0} = \frac{-47.5 \pm 41.68}{-15}$$

$$t_1 = 0.3883, \qquad t_2 = 5.945$$

$$d_1 = \frac{1}{2\pi}\tan^{-1} t_1 = 0.0589\lambda, \qquad d_2 = \frac{1}{2\pi}\tan^{-1} t_2 = 0.2235\lambda$$

At these points $y = 1 + j1.191$ and $y = 1 - j1.191$.

Step 3: Length of the stub

The line susceptance there is $B = \pm 1.191\, Y_0$ (from the formula for B above). The stub must cancel it: $-j \cot \beta l = \mp j1.191$.

$$l_1 = \frac{1}{2\pi}\tan^{-1}\frac{1}{1.191} = 0.1112\lambda, \qquad l_2 = \frac{1}{2\pi}\left[\pi - \tan^{-1}\frac{1}{1.191}\right] = 0.3888\lambda$$

$$\boxed{d_1 = 0.059\lambda,\ l_1 = 0.111\lambda \quad \text{or} \quad d_2 = 0.224\lambda,\ l_2 = 0.389\lambda}$$

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On the Smith chart:

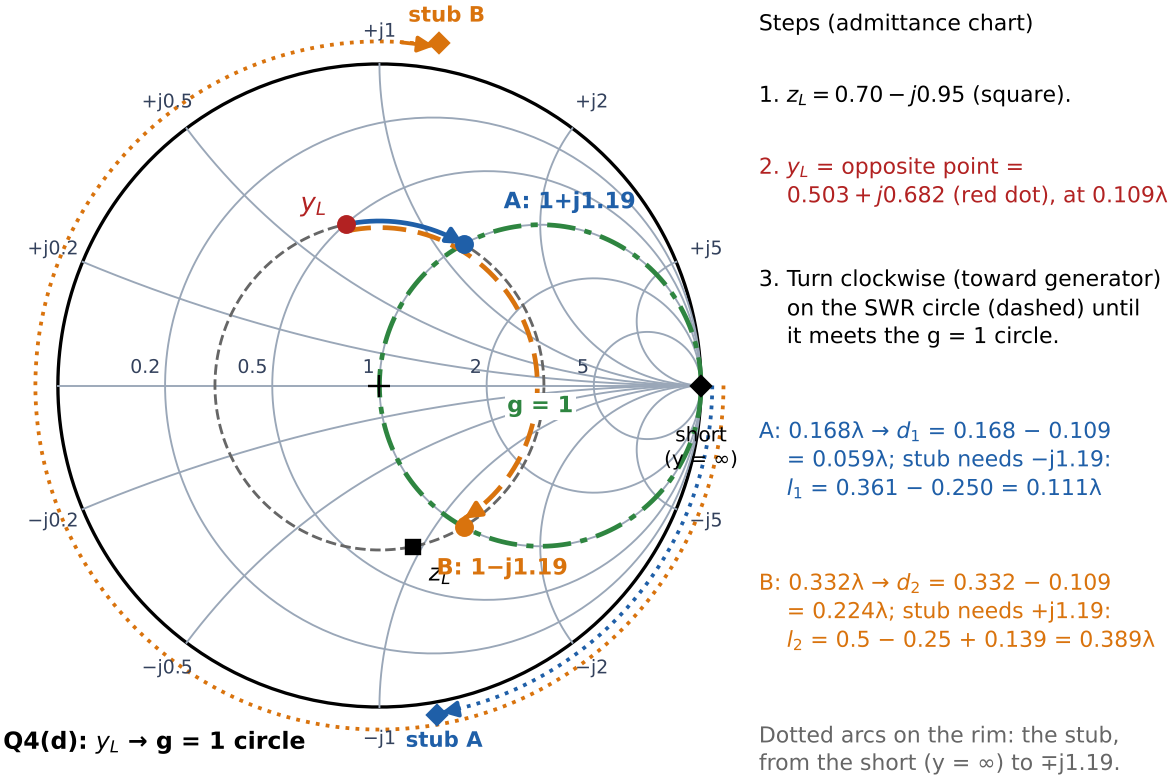


Figure 22: Single-stub matching on the Smith chart

Reading the figure. The square is z_L and the red dot y_L, opposite it on the dashed SWR circle (at 0.109λ on the WTG scale). Turning clockwise meets the dash-dot green $g = 1$ circle at A ($1 + j1.19$, solid blue arc, 0.059λ) and B ($1 - j1.19$, orange arc, 0.224λ). The dotted arcs on the rim are the stubs, from the short ($y = \infty$) to $\mp j1.19$.

Final Answer: the stub is placed 0.059λ from the load with a length of 0.111λ (solution A, shorter and preferred), or 0.224λ from the load with a length of 0.389λ (solution B).

3. Notes and Remarks

- **Check:** at d_1 the line admittance is $1 + j1.1907$ and the stub gives $-j1.1907$: the sum is exactly 1.
- Solution A is preferred: the stub is close to the load, so the stretch of line carrying the high SWR (3.09) is short, giving a wider bandwidth and less loss.

4. Final Concise Answer

(a) (i) $\Gamma = V^-/V^+ = (Z_L - Z_0)/(Z_L + Z_0)$, $|\Gamma| \le 1$, $\Gamma(d) = \Gamma_L e^{-j2\beta d}$. (ii) $S = |V|_{max}/|V|_{min} = (1+|\Gamma|)/(1-|\Gamma|)$, $1 \le S < \infty$, extrema $\lambda/4$ apart. (iii) RF heating (eyes, testes most sensitive), shocks below 10 MHz, risks to fuel and implants; limits on SAR (0.08 W/kg public, 0.4 W/kg occupational, whole body) and on power density (10 / 50 W/m^2 above 2 GHz); keep distance. (iv) A closed metal box resonating at $f_{mnp} = \frac{c}{2\sqrt{\epsilon_r}}\sqrt{(m/a)^2 + (n/b)^2 + (p/d)^2}$, dominant TE_{101}, very high Q; used in filters, oscillators and wavemeters.

(b) A $\lambda/4$ section gives $Z_{in} = Z_1^2/R_L$; with $Z_1 = \sqrt{Z_0 R_L}$, $Z_{in} = Z_0$ (narrow band, resistive load).

(c) $P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$; the object is detected while $P_r \ge P_{min}$, so $R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$; range from $R = ct/2$.

(d) $y_L = 0.503 + j0.682$; stub at $d = 0.059\lambda$, length $l = 0.111\lambda$ (or $d = 0.224\lambda$, $l = 0.389\lambda$).

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5. (a) Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave. Marks: 6
   (b) Find the forward voltage gain of a high-pass filter. Marks: 6
   (c) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. Marks: 7
   (d) Define near field and far field for an antenna. Marks: 7

5.(a) Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave. Marks: 6

1. Theory

Note on the paper

The marks printed for Question 5 (6 + 6 + 7 + 7) add up to 26, while every other question adds up to 25. They are copied as printed.

1. A wave inside a metal pipe (Core Idea)

Inside a rectangular metal pipe (a wide, b high) a wave travels along z by bouncing between the walls. For a **TM** (transverse magnetic) wave the magnetic field lies entirely across the guide ($H_z = 0$), and the electric field has a component along it ($E_z \neq 0$). The walls force the tangential electric field to be zero on them, and that allows only certain patterns, each with its own lowest frequency.

2. What the three quantities mean

- **Phase constant β:** how fast the phase turns along the guide (rad/m).
- **Cutoff frequency f_c:** below it β is imaginary and the wave dies out instead of travelling.
- **Wave impedance Z_{TM}:** the ratio of the transverse electric to the transverse magnetic field, the guide's version of η.

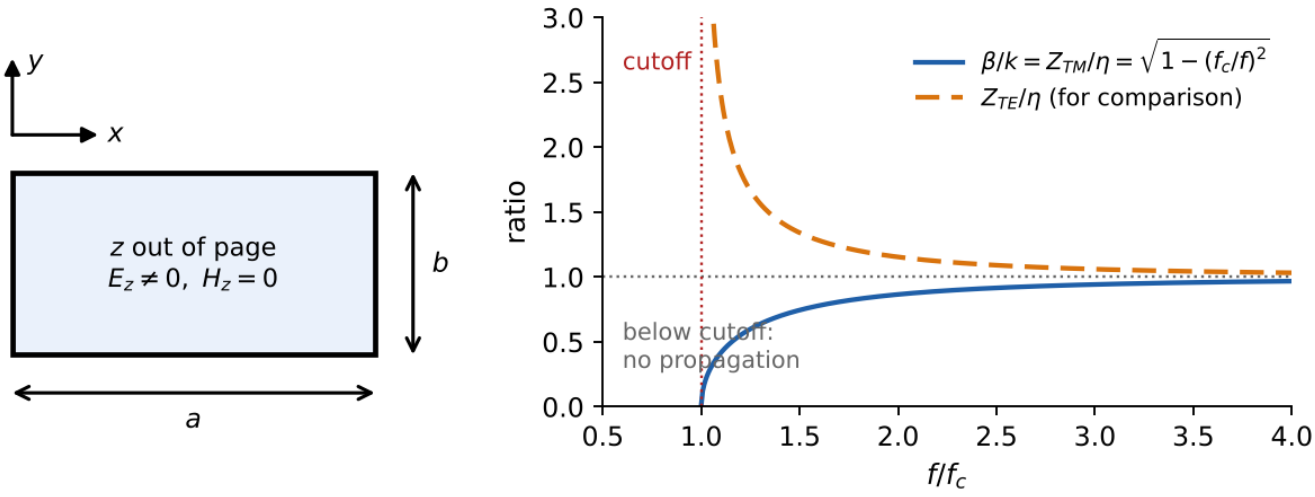


Figure 23: A rectangular guide and the TM curves

Reading the figure. Left: the guide cross-section. Right: $\beta/k = Z_{TM}/\eta$ (solid blue) rises from 0 at cutoff towards 1; the dashed orange curve, Z_{TE}/η, falls from infinity towards 1, for comparison.

2. Answer

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Set-up. Take a rectangular guide, a wide and b high, along z, filled with a medium of μ and ϵ, with no sources inside. All fields vary as $e^{-j\beta z}$, so $\partial/\partial z$ becomes $-j\beta$. A TM wave has $H_z = 0$. Write $k = \omega\sqrt{\mu\epsilon}$.

Step 1: Maxwell's equations

In phasor form,

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \qquad \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$$

Step 2: The wave equation for E_z

Take the curl of the first equation and put in the second:

$$\nabla \times \nabla \times \mathbf{E} = -j\omega\mu\,(\nabla \times \mathbf{H}) = \omega^2\mu\epsilon\,\mathbf{E} = k^2\mathbf{E}$$

Use the identity $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E}$. With no charge inside, $\nabla \cdot \mathbf{E} = 0$, so

$$\nabla^2\mathbf{E} + k^2\mathbf{E} = 0$$

Take its z component, with $E_z(x, y, z) = e_z(x, y)e^{-j\beta z}$, so that $\partial^2/\partial z^2$ becomes $-\beta^2$:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right) e_z = 0, \qquad k_c^2 = k^2 - \beta^2$$

Step 3: Separate the variables

Put $e_z = X(x)Y(y)$ and divide by XY:

$$\frac{X''}{X} + \frac{Y''}{Y} + k_c^2 = 0$$

The first term depends only on x and the second only on y, so each must be a constant. Call them $-k_x^2$ and $-k_y^2$, with $k_x^2 + k_y^2 = k_c^2$. Then $X'' + k_x^2 X = 0$ and $Y'' + k_y^2 Y = 0$, which have the solutions

$$e_z = (A\cos k_x x + B\sin k_x x)(C\cos k_y y + D\sin k_y y)$$

Step 4: Apply the walls

E_z is tangential to all four metal walls, so it must be zero there.

- At $x = 0$: $A = 0$. At $x = a$: $\sin k_x a = 0$, so $k_x = m\pi/a$.
- At $y = 0$: $C = 0$. At $y = b$: $\sin k_y b = 0$, so $k_y = n\pi/b$.

Writing $E_0 = BD$:

$$E_z = E_0 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}\, e^{-j\beta z}, \qquad m, n = 1, 2, 3, \ldots$$

$$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

Step 5: Phase constant

From $k_c^2 = k^2 - \beta^2$:

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Step 6: Cutoff frequency

The wave propagates only while β is real, that is $k > k_c$. At cutoff $k = k_c$ and $\beta = 0$. With $k = 2\pi f\sqrt{\mu\epsilon}$:

$$\boxed{f_{c,mn} = \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

Since $k_c/k = f_c/f$, the phase constant can be written

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$$\boxed{\beta = k\sqrt{1-\left(\frac{k_c}{k}\right)^2} = k\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

Both m and n must be at least 1 (otherwise $E_z = 0$), so the lowest TM mode is TM_{11}.

Step 7: The transverse fields

Write the y component of each curl equation, and the x component of the second, with $\partial/\partial z \to -j\beta$ and $H_z = 0$:

$$-j\beta E_x - \frac{\partial E_z}{\partial x} = -j\omega\mu H_y, \qquad j\beta H_y = j\omega\epsilon E_x$$

The second gives $H_y = (\omega\epsilon/\beta)E_x$. Put this into the first:

$$-j\beta E_x - \frac{\partial E_z}{\partial x} = -j\frac{\omega^2\mu\epsilon}{\beta}E_x = -j\frac{k^2}{\beta}E_x$$

Collect the E_x terms and use $k^2 - \beta^2 = k_c^2$:

$$\frac{\partial E_z}{\partial x} = j\frac{k^2-\beta^2}{\beta}E_x = j\frac{k_c^2}{\beta}E_x$$

$$E_x = \frac{-j\beta}{k_c^2}\frac{\partial E_z}{\partial x}, \qquad H_y = \frac{\omega\epsilon}{\beta}E_x = \frac{-j\omega\epsilon}{k_c^2}\frac{\partial E_z}{\partial x}$$

Step 8: Wave impedance

It is the ratio of the transverse electric field to the transverse magnetic field. Dividing the two results above:

$$Z_{TM} = \frac{E_x}{H_y} = \frac{\beta}{\omega\epsilon}$$

Since $k/(\omega\epsilon) = \omega\sqrt{\mu\epsilon}/(\omega\epsilon) = \sqrt{\mu/\epsilon} = \eta$, this is $\beta\eta/k$. Putting in β from Step 6:

$$\boxed{Z_{TM} = \frac{\beta}{\omega\epsilon} = \frac{\beta\eta}{k} = \eta\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

($-E_y/H_x$ gives the same.) Z_{TM} is real and less than $\eta = \sqrt{\mu/\epsilon}$ above cutoff, and imaginary (no power flow) below it.

Steps to remember

Maxwell's equations → The wave equation for E_z → Separate the variables → Apply the walls → Phase constant → Cutoff frequency → The transverse fields → Wave impedance.

5.(b) Find the forward voltage gain of a high-pass filter. Marks: 6

1. Theory

1. What a high-pass filter does (Core Idea)

A **high-pass filter** lets high frequencies through and blocks low ones. The simplest one is a capacitor in series and a resistor across the output. A capacitor's reactance $1/\omega C$ is huge at low frequency (it blocks) and small at high frequency (it passes). The **forward voltage gain** is the ratio of output to input voltage, $H = V_2/V_1$, a complex number with a size and a phase.

2. The voltage divider

The circuit is just a voltage divider between the capacitor's impedance $1/j\omega C$ and R.

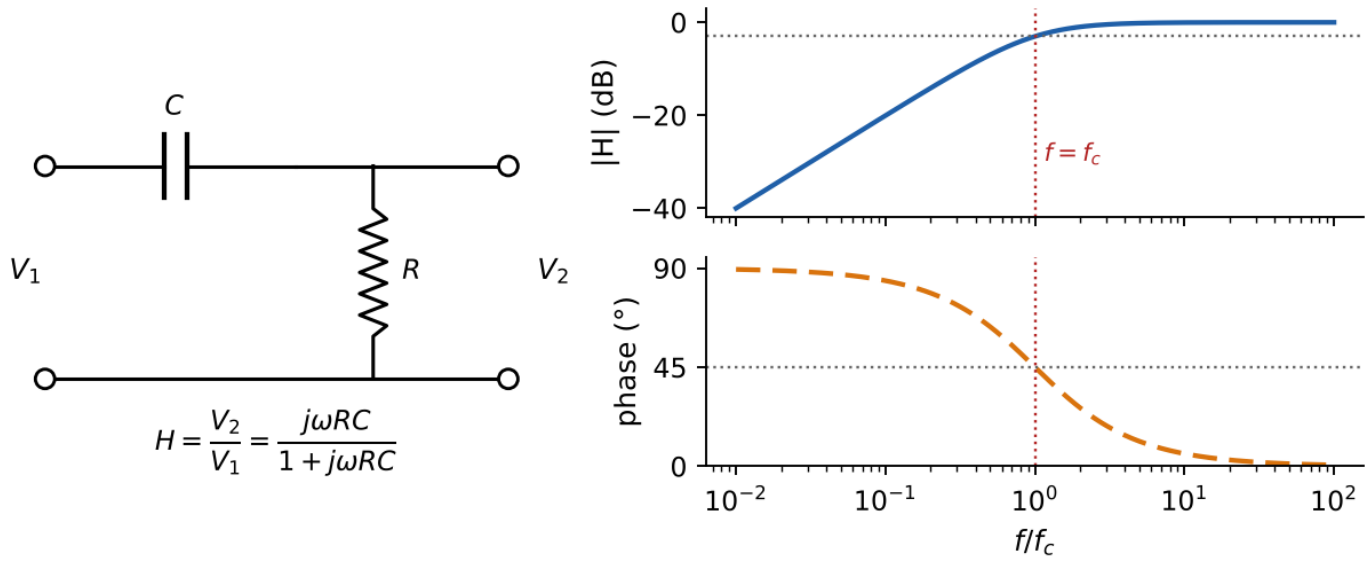


Figure 24: RC high-pass filter and its gain

Reading the figure. Left: series C and shunt R. Right, top: the gain in dB (solid blue) rises at 20 dB per decade and flattens at 0 dB, passing -3 dB at f_c; bottom: the phase (dashed orange) falls from 90° to 0°, through 45° at f_c.

2. Answer

Assumption: the paper gives no circuit, so the standard first-order RC high-pass filter (series C, shunt R, unloaded output) is used.

Step 1: Voltage division

The capacitor has impedance $1/j\omega C$; the output is across R. By voltage division, then multiplying the top and bottom by $j\omega C$:

$$H(\omega) = \frac{V_2}{V_1} = \frac{R}{R + \dfrac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC} = \frac{j(f/f_c)}{1 + j(f/f_c)}, \qquad f_c = \frac{1}{2\pi RC}$$

Step 2: Magnitude and phase

The size of a complex ratio is the ratio of sizes, $|j(f/f_c)| = f/f_c$ and $|1 + j(f/f_c)| = \sqrt{1 + (f/f_c)^2}$; the angle is the difference of angles, 90° for the top and $\tan^{-1}(f/f_c)$ for the bottom:

$$|H| = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}, \qquad \angle H = 90° - \tan^{-1}\frac{f}{f_c}$$

$$\boxed{G_V(\text{dB}) = 20\log_{10}|H| = 20\log_{10}\frac{f/f_c}{\sqrt{1 + (f/f_c)^2}}}$$

Steps to remember

Voltage division $R/(R+1/j\omega C) \to j(f/f_c)/(1+jf/f_c) \to$ magnitude and phase $\to G_V = 20\log|H|$.

Key Points:

- $f \ll f_c$: $|H| \approx f/f_c$, rising at +20 dB/decade, phase $\approx 90°$.

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- $f = f_c$: $|H| = 1/\sqrt{2}$ (−3 dB), phase 45°.
- $f \gg f_c$: $|H| \to 1$ (0 dB), phase $\to 0°$: high frequencies pass unchanged.

3. Notes and Remarks

- **In an RF system** (source and load both Z_0), the forward voltage gain is $S_{21} = 2V_2/V_G$. A single series capacitor between two Z_0 ports is already a high-pass filter: $S_{21} = \dfrac{2Z_0}{2Z_0 + 1/j\omega C} = \dfrac{j\omega(2Z_0C)}{1 + j\omega(2Z_0C)}$, the same form with RC replaced by $2Z_0C$.

5.(c) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. Marks: 7

1. Theory

1. Why use many antennas (Core Idea)

One small antenna has a broad pattern. If we put several identical antennas in a line and feed them together, their waves add in some directions and cancel in others. By choosing the spacing and the phase of each feed we can point a narrow beam where we want. A **uniform** linear array has N identical elements, equally spaced by d, fed with equal amplitudes and a constant phase step δ from one element to the next.

2. The path difference

Seen from a far point at angle θ to the array axis, the wave from each element travels $d\cos\theta$ less than the wave from the one before it, a phase lead $\beta d\cos\theta$. Adding the feed phase step, neighbouring elements differ by

$$\psi = \beta d\cos\theta + \delta$$

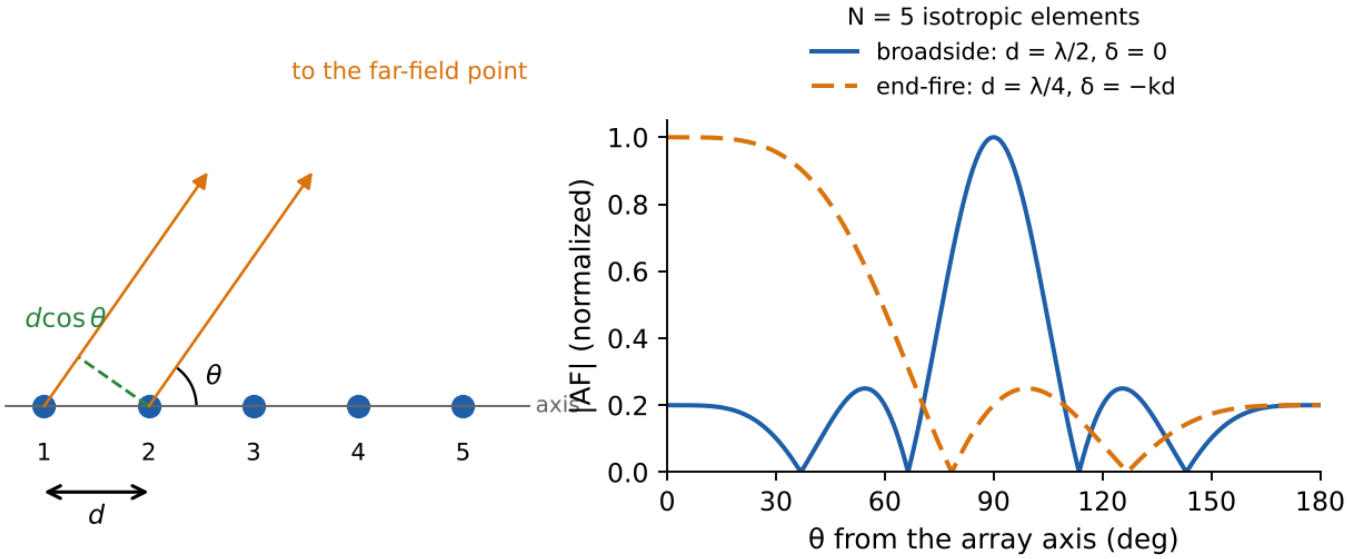


Figure 25: Geometry and array factor of a uniform linear array

Reading the figure. Left: five elements, spacing d, with the extra path $d\cos\theta$ (dashed green) between two neighbours. Right: the array factor of five elements: broadside ($d = \lambda/2$, $\delta = 0$, solid blue) with its main beam at 90°, and end-fire ($d = \lambda/4$, $\delta = -\beta d$, dashed orange) with its beam along the axis at 0°.

2. Answer

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Set-up. Take N identical isotropic elements along a line, a spacing d apart, with equal amplitudes. Each element's feed leads the one before it by the phase δ. A far-away point lies in the direction θ from the line, so the rays from all the elements are parallel.

Step 1: Phase difference between neighbours

The ray from each element has a path shorter by $d\cos\theta$ than the ray from the element before it. That shorter path is a phase lead of $\beta d\cos\theta$, and the feed adds δ. So each element leads the one before it by

$$\psi = \beta d\cos\theta + \delta$$

Step 2: Add the elements

With element 1 as the reference, the total field (the **array factor**) is

$$AF = 1 + e^{j\psi} + e^{j2\psi} + \cdots + e^{j(N-1)\psi}$$

Step 3: Sum the series

This is a geometric series. Call the sum S and let $y = e^{j\psi}$. Then $yS = y + y^2 + \cdots + y^N$. Subtracting, every middle term cancels:

$$S - yS = 1 - y^N \;\Rightarrow\; AF = \frac{1 - e^{jN\psi}}{1 - e^{j\psi}}$$

Step 4: Turn the sum into sines

Take $e^{jN\psi/2}$ out of the top and $e^{j\psi/2}$ out of the bottom, and use $e^{jA} - e^{-jA} = 2j\sin A$:

$$1 - e^{jN\psi} = e^{jN\psi/2}\left(e^{-jN\psi/2} - e^{jN\psi/2}\right) = -2j\, e^{jN\psi/2}\sin\frac{N\psi}{2}$$

$$1 - e^{j\psi} = -2j\, e^{j\psi/2}\sin\frac{\psi}{2}$$

Dividing:

$$AF = e^{j(N-1)\psi/2}\,\frac{\sin(N\psi/2)}{\sin(\psi/2)}$$

Step 5: Normalize

The exponential only sets the phase, which depends on the choice of reference, and has magnitude 1. As $\psi \to 0$, each sine equals its small angle, so the ratio tends to $(N\psi/2)/(\psi/2) = N$. That is the peak value. Dividing by N:

$$\boxed{|AF_n| = \left|\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right|, \qquad \psi = \beta d\cos\theta + \delta}$$

Steps to remember

Phase difference between neighbours → Add the elements → Sum the series → Turn the sum into sines → Normalize.

Main beam direction: the maximum ($|AF_n| = 1$) is at $\psi = 0$:

$$\beta d\cos\theta_m + \delta = 0 \;\Rightarrow\; \theta_m = \cos^{-1}\left(-\frac{\delta}{\beta d}\right)$$

- **Broadside** ($\theta_m = 90°$): $\cos\theta_m = 0$, so $\delta = 0$.
- **End-fire** ($\theta_m = 0°$): $\cos\theta_m = 1$, so $\delta = -\beta d$.
- Changing δ electronically steers the beam (phased array).

Null locations: a null needs the top to be zero while the bottom is not. So $\sin(N\psi/2) = 0$ with $\psi \neq 0$:

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$$\frac{N\psi}{2} = \pm k\pi \;\Rightarrow\; \psi = \pm\frac{2k\pi}{N}, \qquad k = 1, 2, 3, \ldots \quad (k \neq N, 2N, \ldots)$$

Put in $\psi = \beta d\cos\theta_k + \delta$ and solve for θ_k:

$$\theta_k = \cos^{-1}\left[\frac{1}{\beta d}\left(-\delta \pm \frac{2k\pi}{N}\right)\right]$$

For broadside, $\delta = 0$ and $\beta = 2\pi/\lambda$, so $\theta_k = \cos^{-1}\left(\pm\dfrac{k\lambda}{Nd}\right)$.

Width of the main beam, broadside.

First-null beamwidth. The first nulls ($k = 1$) are at $\cos^{-1}(\pm\lambda/Nd)$, on either side of 90°. The beam reaches from 90° down to $\cos^{-1}(\lambda/Nd)$ on each side:

$$\text{FNBW} = 2\left[\frac{\pi}{2} - \cos^{-1}\frac{\lambda}{Nd}\right] = 2\sin^{-1}\frac{\lambda}{Nd} \approx \frac{2\lambda}{Nd}$$

The last step uses $\sin^{-1} y \approx y$ for $Nd \gg \lambda$.

Half-power beamwidth. Half power means $|AF_n| = 1/\sqrt{2} = 0.707$. Near the main beam ψ is small, so $\sin(\psi/2) \approx \psi/2$ and, with $u = N\psi/2$,

$$|AF_n| \approx \frac{\sin u}{u}$$

Solving $\sin u/u = 0.707$ numerically gives $u = 1.391$. A direction Δ away from broadside has $\theta = 90° - \Delta$, so $\psi = \beta d\sin\Delta \approx \beta d\Delta$. Then

$$\frac{N\beta d\Delta}{2} = 1.391 \;\Rightarrow\; \Delta = \frac{2(1.391)}{N(2\pi/\lambda)d} = 0.443\,\frac{\lambda}{Nd}$$

The beam spreads by Δ on each side:

$$\text{HPBW} = 2\Delta \approx 0.886\,\frac{\lambda}{Nd}\ \text{rad}$$

Width of the main beam, end-fire. With $\delta = -\beta d$, $\psi = \beta d(\cos\theta - 1)$. The first null has $\psi = -2\pi/N$:

$$\cos\theta_1 = 1 - \frac{\lambda}{Nd}$$

The beam is centred on $\theta = 0$ and reaches θ_1 on each side, so $\text{FNBW} = 2\theta_1 = 2\cos^{-1}\left(1 - \dfrac{\lambda}{Nd}\right)$. For small θ_1, $\cos\theta_1 \approx 1 - \theta_1^2/2$, which gives $\theta_1 \approx \sqrt{2\lambda/Nd}$:

$$\text{FNBW} \approx 2\sqrt{\frac{2\lambda}{Nd}}$$

The beam narrows as the array length Nd grows.

Example

$N = 5$, $d = \lambda/2$, broadside: nulls at $\cos^{-1}(\pm 0.4) = 66.4°, 113.6°$ and $\cos^{-1}(\pm 0.8) = 36.9°, 143.1°$; FNBW = 47.2°; HPBW = 20.4°; first side lobe 0.25 (−12 dB).

Side lobes and grating lobes: side lobes lie between the nulls, the first about −13 dB for a long array. To avoid extra main beams (grating lobes), $d < \lambda$ for broadside and $d < \lambda/2$ for end-fire.

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5.(d) Define near field and far field for an antenna. Marks: 7

1. Theory

1. Three kinds of terms (Core Idea)

The simplest antenna is a **Hertzian dipole**: a very short wire of length dl carrying a current I. Solving Maxwell's equations for it (from its vector potential $A_z = \frac{\mu_0 I\, dl}{4\pi r} e^{-j\beta r}$, as in 4(a) and 4(b) of the 2020–21 paper) gives

$$H_\phi = \frac{I\, dl\, \sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}, \qquad E_\theta = \frac{\eta\, I\, dl\, \sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}$$

(and a radial E_r with only $1/r^2$ and $1/r^3$ terms). Each bracket mixes three kinds of term:

- $1/r^3$, the **electrostatic** term: the field of the charges piling up at the two ends of the wire, like a capacitor.
- $1/r^2$, the **induction** term: the field of the current itself, like an inductor.
- $1/r$, the **radiation** term: the only one that carries power away for good.

Factor out β/r from the bracket of E_θ: it becomes $\left[j + \frac{1}{\beta r} - \frac{j}{(\beta r)^2}\right]$. The three parts have the same size when $\beta r = 1$, i.e. $r = \lambda/2\pi$. Closer in, the $1/r^3$ and $1/r^2$ parts are much larger: they hold energy that flows out and back each cycle. Farther out, only the $1/r$ part remains.

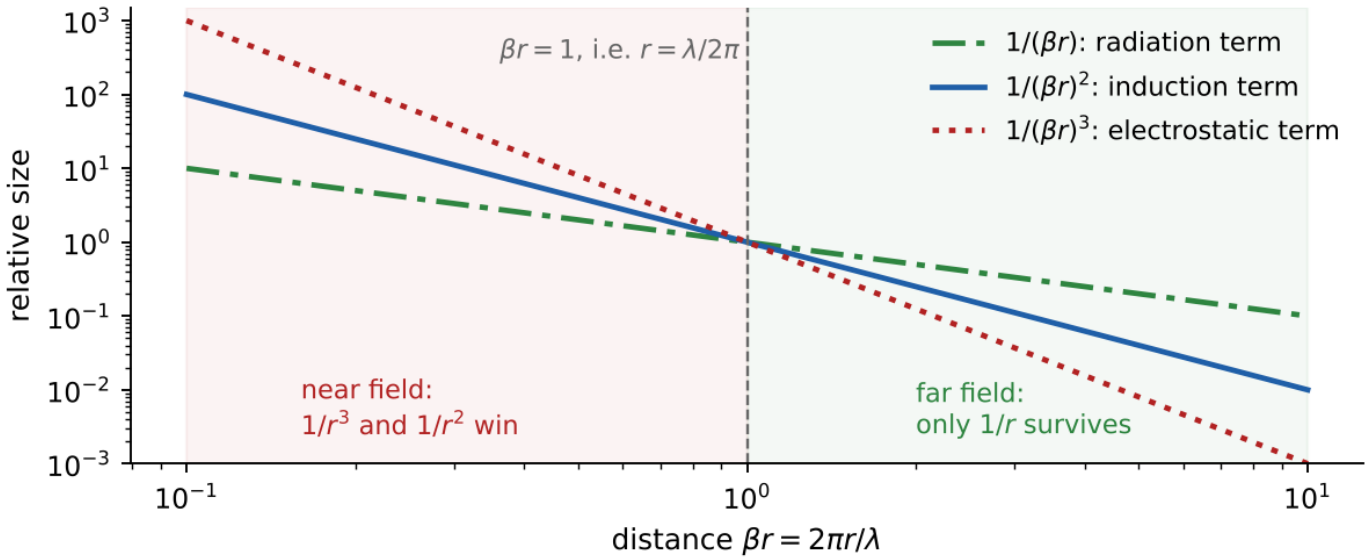


Figure 26: How the three field terms change with distance

Reading the figure. On a log scale, the dotted red $1/r^3$ term and the solid blue $1/r^2$ term are largest close in; the dash-dot green $1/r$ (radiation) term wins beyond $\beta r = 1$.

2. Answer

Near field: the region close to the antenna where the reactive (stored-energy) fields, falling as $1/r^2$ and $1/r^3$, are important. Its pattern changes with distance, $E/H \neq \eta$, and E and H are not in phase. It has two parts:

- **Reactive near field:** $r < 0.62\sqrt{D^3/\lambda}$ (for an antenna small compared with λ, $r < \lambda/2\pi$). Stored energy dominates.
- **Radiating near field (Fresnel region):** $0.62\sqrt{D^3/\lambda} \leq r < 2D^2/\lambda$. Radiation dominates, but the angular pattern still depends on r.

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Far field (Fraunhofer region): $r \geq \dfrac{2D^2}{\lambda}$ (and $r \gg D$, $r \gg \lambda$). Only the $1/r$ radiation fields remain: the angular pattern no longer depends on distance, $\mathbf{E} \perp \mathbf{H} \perp \hat{r}$, $E/H = \eta$, and the power density falls as $1/r^2$, like a plane wave locally. Antenna patterns and gain are defined and measured here.

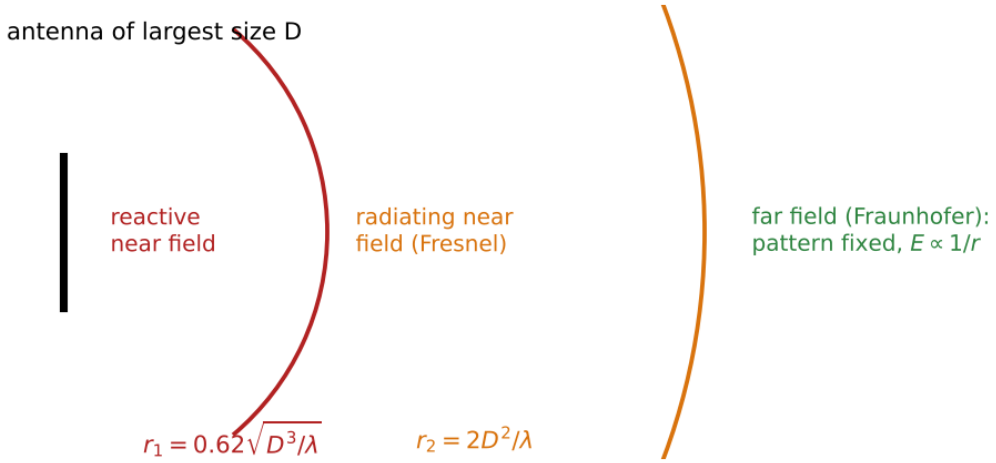


Figure 27: Field regions around an antenna

Reading the figure. The antenna (black bar, largest size D) is at the left. The inner solid red arc at $r_1 = 0.62\sqrt{D^3/\lambda}$ closes the reactive near field; the outer solid orange arc at $r_2 = 2D^2/\lambda$ closes the radiating near field; beyond it lies the far field.

Example

a 3 m dish at 10 GHz ($\lambda = 3$ cm): $r_1 = 0.62\sqrt{27/0.03} = 18.6$ m and $r_2 = 2 \times 9/0.03 = 600$ m; its pattern must be measured more than 600 m away.

	Near field	Far field
Distance	$r < 2D^2/\lambda$	$r \geq 2D^2/\lambda$
Dominant terms	$1/r^3$, $1/r^2$ (stored energy)	$1/r$ (radiation)
Pattern	changes with r	independent of r
E/H	not η, out of phase	η, in phase
Power density	not simply $1/r^2$	$\propto 1/r^2$

Table 4: Near field and far field

3. Final Concise Answer

(a) $E_z = E_0 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b} e^{-j\beta z}$, $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$; $\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - (f_c/f)^2}$; $f_c = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{(m/a)^2 + (n/b)^2}$ (lowest TM_{11}); $Z_{TM} = \beta/\omega\epsilon = \eta\sqrt{1 - (f_c/f)^2}$.

(b) RC high-pass: $H = V_2/V_1 = \frac{j\omega RC}{1+j\omega RC}$, $|H| = \frac{f/f_c}{\sqrt{1+(f/f_c)^2}}$, $\angle H = 90° - \tan^{-1}(f/f_c)$, $f_c = 1/2\pi RC$ (-3 dB, 45°).

(c) $|AF_n| = \left|\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right|$, $\psi = \beta d\cos\theta + \delta$; main beam at $\cos\theta_m = -\delta/\beta d$ (broadside $\delta = 0$, end-fire $\delta = -\beta d$); nulls at $\psi = \pm 2k\pi/N$; broadside FNBW $\approx 2\lambda/Nd$, HPBW $\approx 0.886\lambda/Nd$.

(d) Near field: $r < 2D^2/\lambda$, reactive $1/r^2$, $1/r^3$ terms, pattern depends on r (reactive part $r < 0.62\sqrt{D^3/\lambda}$). Far field: $r \geq 2D^2/\lambda$, only $1/r$ terms, fixed pattern, $E/H = \eta$, $S \propto 1/r^2$.

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6. (a) Explain how radio frequency affects the inductor and capacitor reactance. Marks: 6
   (b) From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient. Marks: 9
   (c) Briefly describe input impedance for open circuit termination and short circuit termination. Marks: 4
   (d) The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L. Marks: 7

6.(a) Explain how radio frequency affects the inductor and capacitor reactance. Marks: 6

1. Theory

Note on the paper

The marks printed for Question 6 (6 + 9 + 4 + 7) add up to 26, while Questions 1 to 4 add up to 25. They are copied as printed.

1. Reactance depends on frequency (Core Idea)

An ideal inductor has reactance $X_L = \omega L$ and an ideal capacitor $X_C = 1/\omega C$. A 10 nH inductor is 0.06 Ω at 1 MHz (almost a wire) but 63 Ω at 1 GHz; a 10 pF capacitor is 16 kΩ at 1 MHz (almost open) but 16 Ω at 1 GHz. So at radio frequencies tiny inductances and capacitances, which we ignore at low frequency, become important.

2. No component is ideal

Every real component has extra (**parasitic**) parts: the leads of a capacitor are short wires with inductance; the turns of an inductor sit next to each other and form small capacitors; every conductor has resistance that grows with frequency (**skin effect**: the current crowds into a thin layer at the surface).

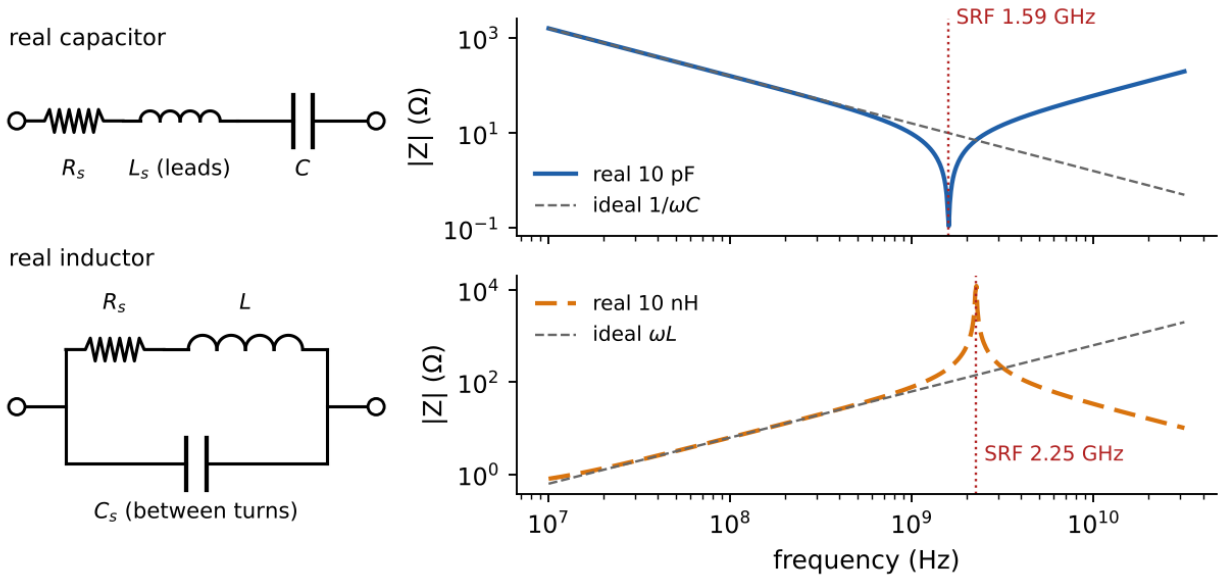


Figure 28: Real L and C at radio frequencies

Reading the figure. Left: the equivalent circuits of a real capacitor (R_s, lead inductance L_s and C in series) and of a real inductor (R_s and L in parallel with the winding capacitance C_s). Right: $|Z|$ against frequency. The real 10 pF capacitor (solid blue) follows the ideal $1/\omega C$ (dashed grey) until a dip at its self-resonance (1.59 GHz), then rises like an inductor. The real 10 nH inductor (dashed

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orange) follows ωL until a peak at 2.25 GHz, then falls like a capacitor.

2. Answer

Capacitor:

- Ideal: $X_C = \dfrac{1}{2\pi f C}$, falling as the frequency rises.
- At RF the lead inductance L_s is in series with C: $X = \omega L_s - \dfrac{1}{\omega C}$. The net reactance falls to zero at the **series self-resonant frequency** $f_r = \dfrac{1}{2\pi\sqrt{L_s C}}$ (1.59 GHz for 10 pF with 1 nH of leads). Above f_r the "capacitor" behaves as an **inductor**.
- Dielectric losses and skin effect add a resistance that grows with frequency.

Inductor:

- Ideal: $X_L = 2\pi f L$, rising with frequency.
- At RF the capacitance C_s between turns is in parallel with L. The impedance rises sharply to a maximum at the **parallel self-resonant frequency** $f_r = \dfrac{1}{2\pi\sqrt{L C_s}}$ (2.25 GHz for 10 nH with 0.5 pF). Above f_r the "inductor" behaves as a **capacitor**.
- The skin effect raises the wire resistance ($R \propto \sqrt{f}$), so the quality factor $Q = \omega L/R$ is limited.

Consequences: every component must be used well below its self-resonant frequency; leads are kept as short as possible (surface-mount chips); and above a few GHz lumped L and C are replaced by distributed elements (short lengths of transmission line, stubs).

6.(b) From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient. Marks: 9

1. Theory

1. The general equation of a line (Core Idea)

A transmission line is two conductors whose length is comparable to the wavelength, so the voltage and current change along it. We cut it into short pieces of length Δz. Each piece has a little series resistance and inductance (the wires) and a little shunt conductance and capacitance (the insulation between them): $R\Delta z$, $L\Delta z$, $G\Delta z$, $C\Delta z$, where R, L, G, C are per metre.

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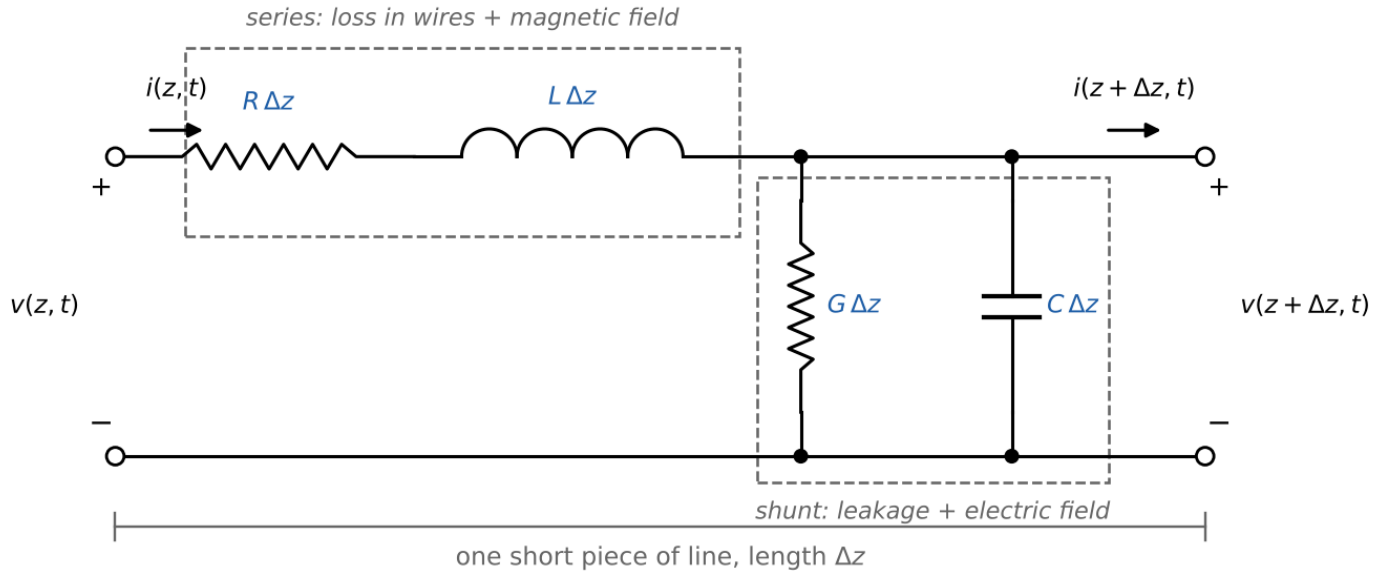


Figure 29: Lumped-element model of a short piece of line

Reading the figure. The voltage $v(z,t)$ and current $i(z,t)$ enter on the left. The series part (top dashed box) drops some voltage, and the shunt part (bottom right box) takes some current, so on the right we get slightly different values $v(z+\Delta z,t)$ and $i(z+\Delta z,t)$.

2. The telegrapher's equations

With phasors ($\partial/\partial t \to j\omega$), Kirchhoff's voltage law around the piece and Kirchhoff's current law at its right-hand node give

$$V(z) - (R + j\omega L)\Delta z\, I(z) - V(z+\Delta z) = 0, \qquad I(z) - (G + j\omega C)\Delta z\, V(z+\Delta z) - I(z+\Delta z) = 0$$

Dividing by Δz and letting $\Delta z \to 0$:

$$\frac{dV}{dz} = -(R + j\omega L)\, I, \qquad \frac{dI}{dz} = -(G + j\omega C)\, V$$

3. The wave equation and its solution

Differentiate the first equation and substitute the second:

$$\frac{d^2V}{dz^2} = -(R + j\omega L)\frac{dI}{dz} = (R + j\omega L)(G + j\omega C)\, V = \gamma^2 V, \qquad \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

A function whose second derivative is γ^2 times itself is $e^{\mp\gamma z}$. So the general solution, a wave going in each direction, is the **general equation of the transmission line**:

$$V(z) = V_0^{+} e^{-\gamma z} + V_0^{-} e^{\gamma z}$$

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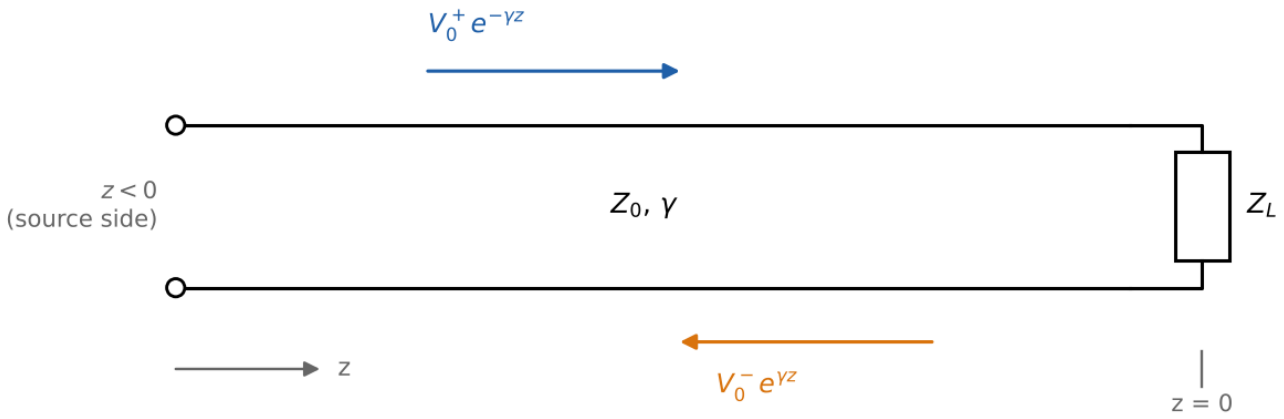


Figure 30: A line with the load at z = 0

Reading the figure. The incident wave $V_0^+ e^{-\gamma z}$ (solid blue) travels towards the load at $z = 0$; the reflected wave $V_0^- e^{\gamma z}$ (orange) travels back towards the source ($z < 0$).

2. Answer

Step 1: Current from the voltage wave

On a line, $dV/dz = -(R + j\omega L)I$ (the first telegrapher's equation, from Kirchhoff's voltage law on a short section, as in the Theory). So $I = -\dfrac{1}{R + j\omega L}\dfrac{dV}{dz}$. Differentiating the general solution:

$$\frac{dV}{dz} = -\gamma V_0^+ e^{-\gamma z} + \gamma V_0^- e^{\gamma z}$$

$$I(z) = \frac{\gamma}{R + j\omega L}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right)$$

Step 2: Characteristic impedance

The ratio of voltage to current in the forward wave alone (or, with a minus sign, in the backward wave) is the characteristic impedance:

$$Z_0 = \frac{V_0^+}{I_0^+} = \frac{R + j\omega L}{\gamma} = \frac{R + j\omega L}{\sqrt{(R + j\omega L)(G + j\omega C)}}$$

$$\boxed{Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}}$$

so that $I(z) = \dfrac{1}{Z_0}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right)$. For a lossless line ($R = G = 0$), $Z_0 = \sqrt{L/C}$, a pure resistance independent of frequency.

Step 3: Reflection coefficient at the load

Put the load Z_L at $z = 0$. There

$$V(0) = V_0^+ + V_0^-, \qquad I(0) = \frac{V_0^+ - V_0^-}{Z_0}, \qquad Z_L = \frac{V(0)}{I(0)} = Z_0\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-}$$

Solving for the ratio of the reflected to the incident wave:

$$Z_L(V_0^+ - V_0^-) = Z_0(V_0^+ + V_0^-) \;\Rightarrow\; V_0^-(Z_L + Z_0) = V_0^+(Z_L - Z_0)$$

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$$\boxed{\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

At any point on the line, $\Gamma(z) = \dfrac{V_0^- e^{\gamma z}}{V_0^+ e^{-\gamma z}} = \Gamma_L e^{2\gamma z}$; a distance l from the load, $\Gamma(-l) = \Gamma_L e^{-2\gamma l}$.

Steps to remember

$I = -(1/(R+j\omega L))\, dV/dz \rightarrow$ differentiate $V_0^{\pm} e^{\mp\gamma z} \rightarrow Z_0 = (R+j\omega L)/\gamma = \sqrt{(R+j\omega L)/(G+j\omega C)}$ $\rightarrow$ at the load $V/I = Z_L \rightarrow$ solve for V_0^-/V_0^+.

Interpretation: $\Gamma_L = 0$ when $Z_L = Z_0$ (matched, no reflection), -1 for a short, $+1$ for an open.

6.(c) Briefly describe input impedance for open circuit termination and short circuit termination. Marks: 4

1. Theory

1. What is input impedance? (Core Idea)

A transmission line carries a forward wave and, if the load is not matched, a reflected wave. The input impedance Z_{in} is the voltage over the current at the sending end. Since both depend on how the two waves add there, Z_{in} depends on the line's length, not only on the load.

2. Where does the formula come from?

With phase constant $\beta = 2\pi/\lambda$ and load reflection coefficient Γ_L, the reflected wave at distance l has travelled there and back, so its phase has changed by $2\beta l$:

$$Z_{in} = Z_0 \frac{1 + \Gamma_L e^{-j2\beta l}}{1 - \Gamma_L e^{-j2\beta l}}$$

Substituting $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$ gives the $\tan \beta l$ form used below.

3. Why are shorted and open lines special?

A short or an open absorbs no power, since $|\Gamma_L| = 1$. So a lossless line ending in either one cannot take power either, and its input must be purely reactive. As l changes, it behaves like an inductor or a capacitor. This is why short lengths of line are used as circuit elements at microwave frequencies.

2. Answer

The input impedance of a lossless line of length l with load Z_L is $Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$ (derived in 3(c)).

Short circuit ($Z_L = 0$):

$$Z_{sc} = jZ_0 \tan \beta l$$

Open circuit ($Z_L \rightarrow \infty$):

$$Z_{oc} = -jZ_0 \cot \beta l$$

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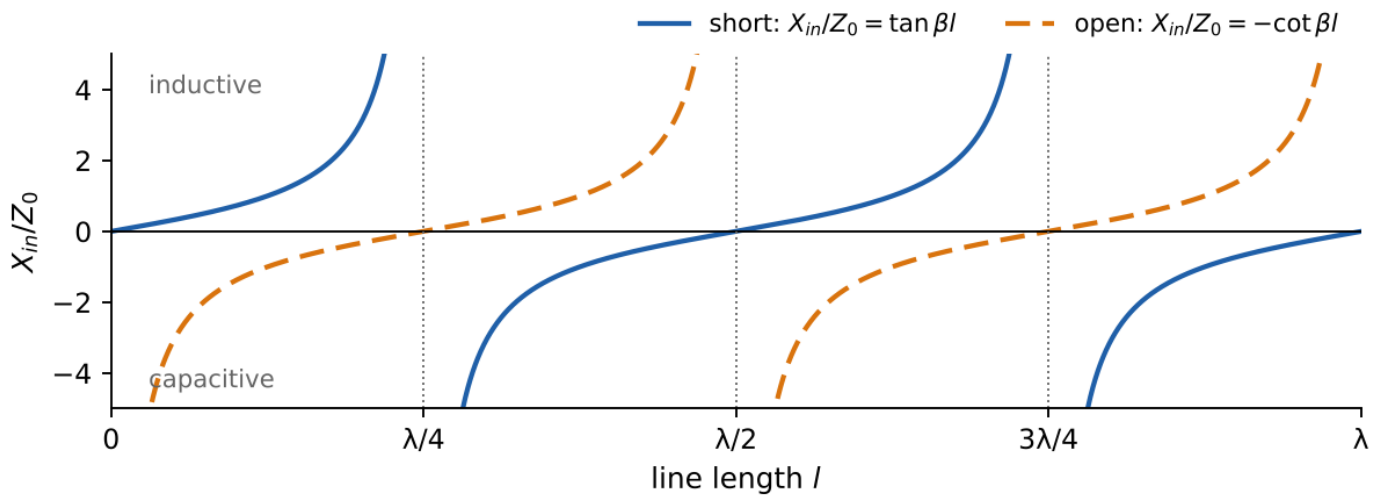


Figure 31: Input reactance of short- and open-circuited lines

Reading the figure. The solid blue curve is the short-circuited line, $\tan \beta l$; the dashed orange curve the open-circuited line, $-\cot \beta l$. Each is a pure reactance that repeats every $\lambda/2$.

Key Points:

- Both are **pure reactances**, taking every value from $-\infty$ to $+\infty$ as l changes: a short stub less than $\lambda/4$ long is an inductor, an open stub less than $\lambda/4$ a capacitor (used as stubs in matching, 4(d)).
- At $l = \lambda/4$ the short looks open and the open looks short; at $l = \lambda/2$ each looks like itself.
- With losses: $Z_{sc} = Z_0 \tanh \gamma l$, $Z_{oc} = Z_0 \coth \gamma l$ (used in 2(d)), and $Z_0 = \sqrt{Z_{oc} Z_{sc}}$, which is how Z_0 and γ of a line are measured.

6.(d) The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L. Marks: 7

1. Theory

1. What the measurements tell us (Core Idea)

- The **SWR** gives the size of the reflection: $|\Gamma| = \dfrac{S-1}{S+1}$.
- **Two successive minima** are half a wavelength apart, so their spacing gives λ.
- At a **voltage minimum** the incident and reflected waves are exactly opposite in phase. Knowing where the first minimum is tells us the phase of Γ.

2. Why the minimum gives the angle

At a distance d from the load, $\Gamma(d) = |\Gamma| e^{j(\theta_\Gamma - 2\beta d)}$. A minimum is where this angle is $-\pi$, so $\theta_\Gamma - 2\beta d_{min} = -\pi$:

$$\theta_\Gamma = 2\beta d_{min} - \pi$$

At the minimum the impedance is real and equal to Z_0/S, its smallest value.

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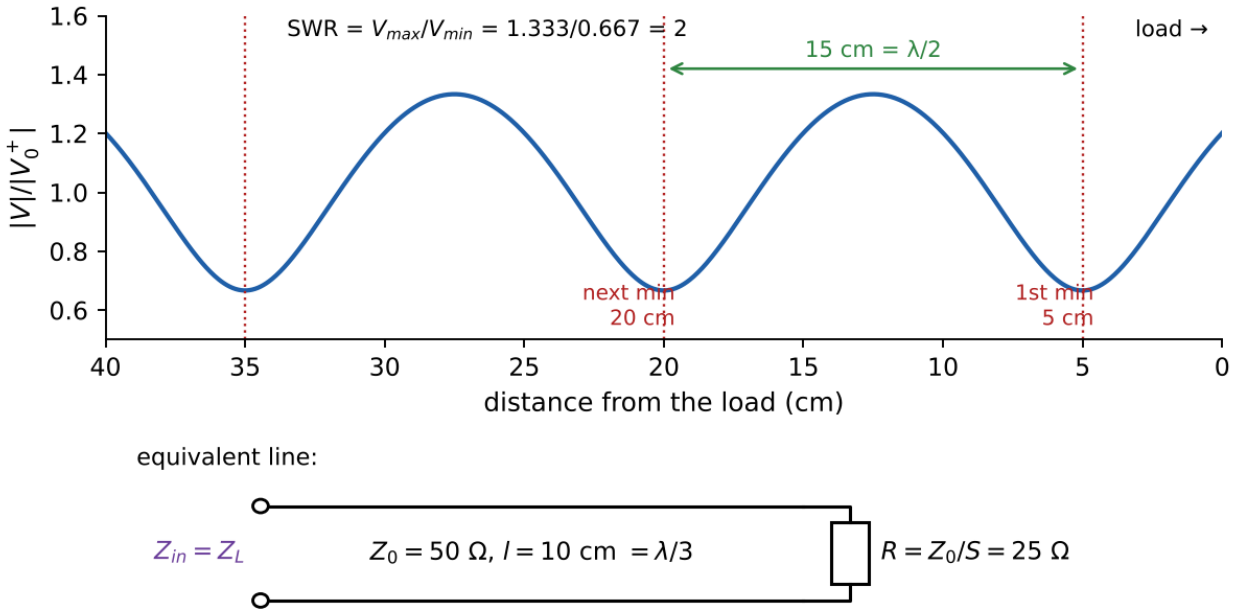


Figure 32: The measured standing wave and the equivalent line

Reading the figure. Top: $|V|$ along the line for this load (solid blue), with the load at the right; the dotted red lines mark the minima at 5 cm and 20 cm, 15 cm ($\lambda/2$) apart. Bottom: the equivalent line, 10 cm long and terminated in 25 Ω, whose input impedance equals Z_L.

2. Answer

Given:

- $Z_0 = 50\ \Omega$, $S = 2$
- Spacing of successive minima = 15 cm, so $\lambda/2 = 15$ cm, $\lambda = 30$ cm and $\beta = 2\pi/\lambda = 20.94$ rad/m
- First minimum at $d_{min} = 5$ cm from the load

Step 1: Reflection coefficient

$$|\Gamma| = \frac{S-1}{S+1} = \frac{2-1}{2+1} = 0.333$$

$$\theta_\Gamma = 2\beta d_{min} - \pi = 2 \times \frac{2\pi}{30} \times 5 - \pi = \frac{2\pi}{3} - \pi = -\frac{\pi}{3} = -60°$$

$$\boxed{\Gamma_L = 0.333\angle -60° = 0.167 - j0.289}$$

Step 2: Load impedance

$$Z_L = Z_0 \frac{1+\Gamma_L}{1-\Gamma_L} = 50 \times \frac{1.167 - j0.289}{0.833 + j0.289} = 57.14 - j37.12\ \Omega$$

$$\boxed{Z_L = 57.1 - j37.1\ \Omega = 68.1\angle -33.0°\ \Omega}$$

Step 3: Equivalent line

Moving $d_{min} = 5$ cm from the load towards the generator brings us to the minimum, where the impedance is the resistance $Z_0/S = 25\ \Omega$. Going on for another $\lambda/2 - d_{min}$ completes half a wavelength, where the impedance is Z_L again. So a line of 25 Ω termination and length $\lambda/2 - d_{min}$ has Z_L at its input:

$$R = \frac{Z_0}{S} = \frac{50}{2} = 25\ \Omega, \qquad l = \frac{\lambda}{2} - d_{min} = 15 - 5 = 10\ \text{cm} = \frac{\lambda}{3}$$

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$$\boxed{l = 10 \text{ cm } (\lambda/3), \qquad R = 25\ \Omega}$$

Check: $Z_{in} = 50\dfrac{25 + j50\tan(120°)}{50 + j25\tan(120°)} = 57.14 - j37.12\ \Omega = Z_L$. ✔

Final Answer: $\Gamma_L = 0.333\angle -60°$; $Z_L = 57.1 - j37.1\ \Omega$; an equivalent line 10 cm ($\lambda/3$) long terminated in $R = 25\ \Omega$ (or, using the voltage maximum, 2.5 cm terminated in $Z_0 S = 100\ \Omega$).

3. Notes and Remarks

- The same result in one formula: $Z_L = Z_0\dfrac{1 - jS\tan\beta d_{min}}{S - j\tan\beta d_{min}} = 50\dfrac{1 - j3.464}{2 - j1.732} = 57.14 - j37.12\ \Omega$.
- The voltage maximum is $\lambda/4 = 7.5$ cm beyond the minimum (12.5 cm from the load), where $Z = Z_0 S = 100\ \Omega$; a line of $15 - 12.5 = 2.5$ cm terminated in 100 Ω also has input Z_L.

4. Final Concise Answer

(a) $X_L = \omega L$ grows and $X_C = 1/\omega C$ falls with frequency, so at RF small parasitics matter: a capacitor's lead inductance makes it series-resonant at $1/2\pi\sqrt{L_s C}$ and inductive above; an inductor's winding capacitance makes it parallel-resonant at $1/2\pi\sqrt{LC_s}$ and capacitive above; skin effect raises R. Use parts below their self-resonance, or distributed elements.

(b) From $V = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$ and $dV/dz = -(R + j\omega L)I$: $I = \frac{1}{Z_0}(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z})$ with $Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$ ($\sqrt{L/C}$ lossless); at the load $z = 0$, $\Gamma_L = V_0^-/V_0^+ = \frac{Z_L - Z_0}{Z_L + Z_0}$, and $\Gamma(z) = \Gamma_L e^{2\gamma z}$.

(c) $Z_{sc} = jZ_0\tan\beta l$, $Z_{oc} = -jZ_0\cot\beta l$: pure reactances, inductive/capacitive below $\lambda/4$, swapping every $\lambda/4$; lossy: $Z_0\tanh\gamma l$, $Z_0\coth\gamma l$; $Z_0 = \sqrt{Z_{oc}Z_{sc}}$.

(d) $\lambda = 30$ cm, $|\Gamma| = 1/3$, $\theta_\Gamma = 2\beta d_{min} - \pi = -60°$: $\Gamma_L = 0.333\angle -60°$, $Z_L = 57.1 - j37.1\ \Omega$; equivalent line $l = \lambda/2 - d_{min} = 10$ cm terminated in $R = Z_0/S = 25\ \Omega$.

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FORMULA SHEET

EEE475

RF and Microwave Engineering

Formula Sheet

Session 2019–20 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2023

TOPICS

1 Transmission lines — 18 formulas

2 Smith chart and matching — 7 formulas

3 Waveguides and cavities — 9 formulas

4 RF components and filters — 6 formulas

5 S-parameters and amplifiers — 10 formulas

6 Antennas — 10 formulas

7 Arrays and field regions — 7 formulas

8 Remote sensing, radar and safety — 10 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

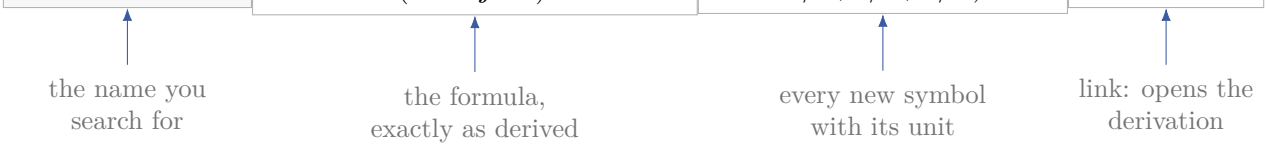
October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

Telegrapher's equations	$\frac{dV}{dz} = -(R + j\omega L)\, I,\ \frac{dI}{dz} = -(G + j\omega C)\, V$	R, L, G, C = line constants per metre (Ω/m, H/m, S/m, F/m)	6(b)



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Transmission lines 4

2 Smith chart and matching 5

3 Waveguides and cavities 5

4 RF components and filters 6

5 S-parameters and amplifiers 7

6 Antennas 7

7 Arrays and field regions 8

8 Remote sensing, radar and safety 8

9 Key formulas 10

10 Symbol index 12

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Conventions (Pozar for lines, waveguides and S-parameters; Ludwig and Bretchko for amplifiers and RF components; Kraus for antennas): phasors with $e^{j\omega t}$; the load is at $z = 0$ and the source at $z < 0$; d is a distance from the load towards the generator; z, y are normalized to Z_0; $\eta_0 = 120\pi\ \Omega$, $c = 3 \times 10^8$ m/s, $k = 1.38 \times 10^{-23}$ J/K; θ is measured from the array or dipole axis. References point to the solutions of the 2019–20 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Transmission lines:

Title	Equation	Notation	Ref.
Telegrapher's equations	$\frac{dV}{dz} = -(R + j\omega L)\, I,\ \frac{dI}{dz} = -(G + j\omega C)\, V$	R, L, G, C = line constants per metre (Ω/m, H/m, S/m, F/m)	6(b)
★ **General solution**	$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$	V_0^+, V_0^- = incident and reflected amplitudes (V)	6(b)
Line current	$I(z) = \frac{1}{Z_0}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right)$	forward minus backward wave, divided by Z_0	6(b)
Propagation constant	$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$	α = attenuation (Np/m) β = phase constant (rad/m)	6(b)
★ **Characteristic impedance**	$Z_0 = \frac{R + j\omega L}{\gamma} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$	Z_0 = characteristic impedance (Ω); lossless: $\sqrt{L/C}$	6(b)
★ **Reflection coefficient**	$\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$	Γ_L = reflection coefficient at the load Z_L = load impedance (Ω)	6(b)
Along the line	$\Gamma(z) = \Gamma_L e^{2\gamma z},\ \Gamma(d) = \Gamma_L e^{-j2\beta d}$	d = distance from the load (m), lossless	4(a)
Return loss	$RL = -20 \log_{10}\lvert\Gamma\rvert$ dB	$\lvert\Gamma\rvert^2$ = fraction of power reflected	4(a)
★ **Standing wave ratio**	$S = \frac{\lvert V\rvert_{max}}{\lvert V\rvert_{min}} = \frac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert},\ \lvert\Gamma\rvert = \frac{S - 1}{S + 1}$	S = SWR, $1 \le S < \infty$	4(a)
★ **Input impedance**	$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	l = line length (m), lossless	3(c)
Input reflection	$\Gamma_{in} = \Gamma_L e^{-j2\beta l}$	from the flow graph of a loaded line	3(c)
Lossy input impedance	$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l}$	$\tanh\gamma l$ replaces $j\tan\beta l$	2(d)
★ **Short and open lines**	$Z_{sc} = jZ_0 \tan\beta l,\ Z_{oc} = -jZ_0 \cot\beta l$	lossy: $Z_0 \tanh\gamma l$, $Z_0 \coth\gamma l$	6(c)
Z_0 from two measurements	$Z_0 = \sqrt{Z_{oc} Z_{sc}}$	Z_{oc}, Z_{sc} = measured input impedances (Ω)	6(c)
γ from a short	$\gamma l = \frac{1}{2} \ln \frac{1 + T}{1 - T},\ T = \tanh\gamma l = \frac{Z_{sc}}{Z_0}$	βl found only up to $n\pi$	2(d)
Phase from the minimum	$\theta_\Gamma = 2\beta d_{min} - \pi$	d_{min} = distance of the first voltage minimum (m)	6(d)

Title	Equation	Notation	Ref.
Impedance at extrema	$Z_{min} = \frac{Z_0}{S},\ Z_{max} = Z_0 S$	real; at the voltage minimum and maximum	6(d)
★ **Load from SWR**	$Z_L = Z_0 \frac{1 - jS\tan\beta d_{min}}{S - j\tan\beta d_{min}}$	minima $\lambda/2$ apart	6(d)

Table 1: Transmission lines

2. Smith chart and matching:

Title	Equation	Notation	Ref.
Impedance and Γ	$z = r + jx = \frac{1+\Gamma}{1-\Gamma}$	$\Gamma = u + jv$; r, x = normalized resistance and reactance	2(a)
★ **Constant-r circle**	$\left(u - \frac{r}{1+r}\right)^2 + v^2 = \left(\frac{1}{1+r}\right)^2$	centre $(r/(1+r), 0)$, radius $1/(1+r)$	2(a)
★ **Constant-x circle**	$(u-1)^2 + \left(v - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$	centre $(1, 1/x)$, radius $1/\lvert x\rvert$	2(b)
★ **Quarter-wave transformer**	$Z_{in} = \frac{Z_1^2}{R_L},\ Z_1 = \sqrt{Z_0 R_L}$	Z_1 = impedance of the $\lambda/4$ section (Ω) R_L = resistive load (Ω)	4(b)
Stub position	$t = \tan\beta d = \frac{X_L \pm \sqrt{R_L\left[(Z_0 - R_L)^2 + X_L^2\right]/Z_0}}{R_L - Z_0}$	R_L, X_L = load resistance and reactance (Ω)	4(d)
Susceptance to cancel	$B = \frac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0\left[R_L^2 + (X_L + Z_0 t)^2\right]}$	B = line susceptance at the stub (S)	4(d)
★ **Short-circuited stub**	$y_{stub} = -j\cot\beta l,\ \frac{l}{\lambda} = \frac{1}{2\pi}\tan^{-1}\frac{Y_0}{B}$	$Y_0 = 1/Z_0$; add $\lambda/2$ if negative	4(d)

Table 2: Smith chart and matching

3. Waveguides and cavities:

Title	Equation	Notation	Ref.
TM field	$E_z = E_0 \sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}e^{-j\beta z}$	a, b = guide width and height (m) $m, n \geq 1$	5(a)
Cutoff wave number	$k_c = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2},\ k_c^2 = k^2 - \beta^2$	$k = \omega\sqrt{\mu\epsilon}$ = wave number (rad/m)	5(a)
★ **Phase constant**	$\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$	real above cutoff only	5(a)

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Title	Equation	Notation	Ref.
★ **Cutoff frequency**	$f_{c,mn} = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$	lowest TM mode: TM_{11}	5(a)
★ **TM wave impedance**	$Z_{TM} = \frac{E_x}{H_y} = \frac{\beta}{\omega\epsilon} = \eta\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$	$\eta = \sqrt{\mu/\epsilon}$ (Ω)	5(a)
TE wave impedance	$Z_{TE} = \frac{k\eta}{\beta}$	above η	2(c)
Circular guide cutoff	$f_c = \frac{p\,c}{2\pi a\sqrt{\epsilon_r}}$	a = radius (m); TE: $p = p'_{nm}$ (TE_{11}: 1.841), TM: $p = p_{nm}$ (TM_{01}: 2.405)	2(c)
★ **Cavity resonance**	$f_{mnp} = \frac{c}{2\sqrt{\mu_r\epsilon_r}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$	d = cavity length (m); dominant TE_{101}	4(a)
Quality factor	$Q = \omega\,\frac{\text{energy stored}}{\text{power lost}}$	thousands to tens of thousands for a cavity	4(a)

Table 3: Waveguides and cavities

4. RF components and filters:

Title	Equation	Notation	Ref.
Ideal reactances	$X_L = 2\pi fL,\ X_C = \frac{1}{2\pi fC}$	f = frequency (Hz)	6(a)
★ **Capacitor self-resonance**	$f_r = \frac{1}{2\pi\sqrt{L_sC}}$	L_s = lead inductance (H); inductive above f_r	6(a)
★ **Inductor self-resonance**	$f_r = \frac{1}{2\pi\sqrt{LC_s}}$	C_s = winding capacitance (F); capacitive above f_r	6(a)
★ **High-pass gain**	$H = \frac{V_2}{V_1} = \frac{j\omega RC}{1 + j\omega RC},\ f_c = \frac{1}{2\pi RC}$	V_1, V_2 = input and output voltages	5(b)
Gain and phase	$\lvert H\rvert = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}},\ \angle H = 90° - \tan^{-1}\frac{f}{f_c}$	−3 dB and 45° at f_c	5(b)
Series C in a Z_0 system	$S_{21} = \frac{j\omega(2Z_0C)}{1 + j\omega(2Z_0C)}$	forward voltage gain between Z_0 ports	5(b)

Table 4: RF components and filters

5. S-parameters and amplifiers:

Title	Equation	Notation	Ref.
Power waves	$a_n = \frac{V_n^+}{\sqrt{Z_0}},\ b_n = \frac{V_n^-}{\sqrt{Z_0}}$	a_n, b_n = incident and outgoing waves at port n ($\sqrt{\text{W}}$)	3(a)
★ **S-parameters**	$[b] = [S][a],\ S_{ij} = \left.\frac{b_i}{a_j}\right\|_{a_k=0,\, k\neq j}$	other ports matched	3(a)
Ideal circulator	$S_{21} = S_{32} = S_{13} = 1$, all other $S_{ij} = 0$	$b_2 = a_1$, $b_3 = a_2$, $b_1 = a_3$; matched, lossless, non-reciprocal	3(a)
Flow graph with the source	$a_1 = \frac{b_s}{1 - \Gamma_S \Gamma_L e^{-j2\beta l}}$	b_s = source wave Γ_S = source reflection coefficient	3(c)
Unilateral gain	$G_{TU} = G_S\, G_0\, G_L,\ G_0 = \lvert S_{21}\rvert^2$	$S_{12} \approx 0$	3(b)
★ **Source gain**	$G_S = \frac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$	G_L: same with S_{22}, Γ_L	3(b)
★ **Maximum source gain**	$G_{Smax} = \frac{1}{1 - \lvert S_{11}\rvert^2}$	at $\Gamma_S = S_{11}^*$	3(d)
Normalized gain	$g_s = \frac{G_S}{G_{Smax}} = G_S(1 - \lvert S_{11}\rvert^2)$	$0 \le g_s \le 1$	3(b)
★ **Gain-circle centre**	$d_{gs} = \frac{g_s S_{11}^*}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	d_{gs} = centre in the Γ_S plane	3(b)
★ **Gain-circle radius**	$r_{gs} = \frac{\sqrt{1 - g_s}\,(1 - \lvert S_{11}\rvert^2)}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	r_{gs} = radius	3(b)

Table 5: S-parameters and amplifiers

6. Antennas:

Title	Equation	Notation	Ref.
★ **Directivity**	$D = \frac{U_{max}}{U_{av}} = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi}{\Omega_A}$	U = radiation intensity (W/sr) P_{rad} = radiated power (W)	1(a)
Directivity estimate	$D \approx \frac{41\,253}{\theta_{HP}\,\phi_{HP}}$	θ_{HP}, ϕ_{HP} = half-power beamwidths (deg)	1(a)
Gain	$G = k\,D$	k = radiation efficiency	1(a)
★ **Beam area**	$\Omega_A = \iint P_n(\theta,\phi)\sin\theta\, d\theta\, d\phi \approx \theta_{HP}\,\phi_{HP}$	P_n = normalized power pattern; sr	1(a)
★ **Effective height**	$h_e = \frac{V_{oc}}{E} = \frac{1}{I_0}\int I(z)\, dz$	V_{oc} = open-circuit voltage (V) I_0 = terminal current (A); half-wave dipole λ/π	1(a)
Height and aperture	$A_e = \frac{h_e^2\, Z_0}{4R_r}$	$Z_0 = 377\ \Omega$; R_r = radiation resistance (Ω)	1(a)

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Title	Equation	Notation	Ref.
★ **Effective aperture**	$A_e = \frac{P_r}{S},\ \varepsilon_{ap} = \frac{A_e}{A_p}$	P_r = received power (W) S = power density (W/m^2) A_p = physical aperture (m^2)	1(b)
★ **Aperture and directivity**	$D = \frac{4\pi A_e}{\lambda^2},\ A_e\,\Omega_A = \lambda^2$	λ = wavelength (m)	1(b)
Aperture beamwidth	$\text{HPBW} \approx \frac{51^\circ\,\lambda}{a}$	a = width of a uniform aperture (m); dish $\approx 70^\circ\lambda/D$	1(b)
Radiated power	$P_{rad} = \int_0^{2\pi}\int_0^{\pi} U\,\sin\theta\,d\theta\,d\phi$	$\sin\theta\,d\theta\,d\phi$ = element of solid angle (sr)	1(d)

Table 6: Antennas

7. Arrays and field regions:

Title	Equation	Notation	Ref.
Phase difference	$\psi = \beta d\cos\theta + \delta$	d = element spacing (m) δ = progressive feed phase (rad)	5(c)
★ **Array factor**	$\lvert AF_n\rvert = \left\lvert\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right\rvert$	N = number of elements	5(c)
★ **Main beam**	$\cos\theta_m = -\frac{\delta}{\beta d}$	broadside $\delta = 0$; end-fire $\delta = -\beta d$	5(c)
★ **Nulls**	$\theta_k = \cos^{-1}\left[\frac{1}{\beta d}\left(-\delta \pm \frac{2k\pi}{N}\right)\right]$	$k = 1, 2, \ldots$, not a multiple of N	5(c)
Broadside beamwidths	$\text{FNBW} = 2\left[\frac{\pi}{2} - \cos^{-1}\frac{\lambda}{Nd}\right],\ \text{HPBW} \approx 0.886\,\frac{\lambda}{Nd}$	in radians	5(c)
End-fire beamwidth	$\text{FNBW} = 2\cos^{-1}\left(1 - \frac{\lambda}{Nd}\right)$	wider than broadside for the same Nd	5(c)
★ **Field regions**	$r_1 = 0.62\sqrt{\frac{D^3}{\lambda}},\ r_2 = \frac{2D^2}{\lambda}$	D = largest antenna size (m); far field beyond r_2	5(d)

Table 7: Arrays and field regions

8. Remote sensing, radar and safety:

Title	Equation	Notation	Ref.
Brightness temperature	$T_B = e\,T_{phys},\ e = 1 - \lvert\Gamma\rvert^2$	e = emissivity T_{phys} = physical temperature (K)	1(c)
Antenna temperature	$T_A = \frac{1}{\Omega_A}\iint T_B\,P_n\,d\Omega$	the scene averaged over the beam (K)	1(c)

Title	Equation	Notation	Ref.
Received noise power	$P = k\,T_A\,B$	k = Boltzmann's constant B = bandwidth (Hz)	1(c)
★ **Radiometer resolution**	$\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$	T_{sys} = system noise temperature (K) τ = integration time (s)	1(c)
Density at the target	$S_t = \dfrac{P_t G}{4\pi R^2}$	P_t = transmitted power (W) G = antenna gain R = range (m)	4(c)
★ **Radar equation**	$P_r = \dfrac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$	σ = radar cross-section (m^2)	4(c)
★ **Maximum range**	$R_{max} = \left[\dfrac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$	P_{min} = minimum detectable signal (W)	4(c)
Range from delay	$R = \dfrac{c\,t}{2}$	t = echo delay (s)	4(c)
Specific absorption rate	$\mathrm{SAR} = \dfrac{\sigma_t \lvert E\rvert^2}{\rho}$	σ_t = tissue conductivity (S/m) ρ = tissue density ($\mathrm{kg/m}^3$); W/kg	4(a)
Safe distance	$r = \sqrt{\dfrac{\mathrm{EIRP}}{4\pi S_{lim}}}$	S_{lim} = power-density limit ($\mathrm{W/m}^2$)	4(a)

Table 8: Remote sensing, radar and safety

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9. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
General solution	$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$
Characteristic impedance	$Z_0 = \dfrac{R + j\omega L}{\gamma} = \sqrt{\dfrac{R + j\omega L}{G + j\omega C}}$
Reflection coefficient	$\Gamma_L = \dfrac{V_0^-}{V_0^+} = \dfrac{Z_L - Z_0}{Z_L + Z_0}$
Standing wave ratio	$S = \dfrac{\lvert V\rvert_{max}}{\lvert V\rvert_{min}} = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert},\ \lvert\Gamma\rvert = \dfrac{S-1}{S+1}$
Input impedance	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$
Short and open lines	$Z_{sc} = jZ_0 \tan\beta l,\ Z_{oc} = -jZ_0 \cot\beta l$
Load from SWR	$Z_L = Z_0 \dfrac{1 - jS\tan\beta d_{min}}{S - j\tan\beta d_{min}}$
Constant-r circle	$\left(u - \dfrac{r}{1+r}\right)^2 + v^2 = \left(\dfrac{1}{1+r}\right)^2$
Constant-x circle	$(u-1)^2 + \left(v - \dfrac{1}{x}\right)^2 = \left(\dfrac{1}{x}\right)^2$
Quarter-wave transformer	$Z_{in} = \dfrac{Z_1^2}{R_L},\ Z_1 = \sqrt{Z_0 R_L}$
Short-circuited stub	$y_{stub} = -j\cot\beta l,\ \dfrac{l}{\lambda} = \dfrac{1}{2\pi}\tan^{-1}\dfrac{Y_0}{B}$
Phase constant	$\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$
Cutoff frequency	$f_{c,mn} = \dfrac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$
TM wave impedance	$Z_{TM} = \dfrac{E_x}{H_y} = \dfrac{\beta}{\omega\epsilon} = \eta\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$
Cavity resonance	$f_{mnp} = \dfrac{c}{2\sqrt{\mu_r\epsilon_r}}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2 + \left(\dfrac{p}{d}\right)^2}$
Capacitor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{L_s C}}$
Inductor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{LC_s}}$
High-pass gain	$H = \dfrac{V_2}{V_1} = \dfrac{j\omega RC}{1 + j\omega RC},\ f_c = \dfrac{1}{2\pi RC}$
S-parameters	$[b] = [S][a],\ S_{ij} = \left.\dfrac{b_i}{a_j}\right\rvert_{a_k=0,\, k\neq j}$
Source gain	$G_S = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$

Title	Equation
Maximum source gain	$G_{Smax} = \dfrac{1}{1-\lvert S_{11}\rvert^2}$
Gain-circle centre	$d_{gs} = \dfrac{g_s S_{11}^*}{1-\lvert S_{11}\rvert^2(1-g_s)}$
Gain-circle radius	$r_{gs} = \dfrac{\sqrt{1-g_s}\,(1-\lvert S_{11}\rvert^2)}{1-\lvert S_{11}\rvert^2(1-g_s)}$
Directivity	$D = \dfrac{U_{max}}{U_{av}} = \dfrac{4\pi U_{max}}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$
Beam area	$\Omega_A = \iint P_n(\theta,\phi)\sin\theta\,d\theta\,d\phi \approx \theta_{HP}\,\phi_{HP}$
Effective height	$h_e = \dfrac{V_{oc}}{E} = \dfrac{1}{I_0}\displaystyle\int I(z)\,dz$
Effective aperture	$A_e = \dfrac{P_r}{S},\ \varepsilon_{ap} = \dfrac{A_e}{A_p}$
Aperture and directivity	$D = \dfrac{4\pi A_e}{\lambda^2},\ A_e\,\Omega_A = \lambda^2$
Array factor	$\lvert AF_n\rvert = \left\lvert \dfrac{\sin(N\psi/2)}{N\sin(\psi/2)} \right\rvert$
Main beam	$\cos\theta_m = -\dfrac{\delta}{\beta d}$
Nulls	$\theta_k = \cos^{-1}\left[\dfrac{1}{\beta d}\left(-\delta \pm \dfrac{2k\pi}{N}\right)\right]$
Field regions	$r_1 = 0.62\sqrt{\dfrac{D^3}{\lambda}},\ r_2 = \dfrac{2D^2}{\lambda}$
Radiometer resolution	$\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$
Radar equation	$P_r = \dfrac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$
Maximum range	$R_{max} = \left[\dfrac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$

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10. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
R, L, G, C	line constants per metre (Ω/m, H/m, S/m, F/m)	Telegrapher's equations
V_0^+, V_0^-	incident and reflected amplitudes (V)	General solution
α	attenuation (Np/m)	Propagation constant
β	phase constant (rad/m)	Propagation constant
Z_0	characteristic impedance (Ω); lossless: $\sqrt{L/C}$	Characteristic impedance
Γ_L	reflection coefficient at the load	Reflection coefficient
Z_L	load impedance (Ω)	Reflection coefficient
d	distance from the load (m), lossless	Along the line
$\lvert\Gamma\rvert^2$	fraction of power reflected	Return loss
S	SWR, $1 \le S < \infty$	Standing wave ratio
l	line length (m), lossless	Input impedance
Z_{oc}, Z_{sc}	measured input impedances (Ω)	Z_0 from two measurements
d_{min}	distance of the first voltage minimum (m)	Phase from the minimum
$\Gamma = u + jv$; r, x	normalized resistance and reactance	Impedance and Γ
Z_1	impedance of the $\lambda/4$ section (Ω)	Quarter-wave transformer
R_L	resistive load (Ω)	Quarter-wave transformer
R_L, X_L	load resistance and reactance (Ω)	Stub position
B	line susceptance at the stub (S)	Susceptance to cancel
a, b	guide width and height (m)	TM field
$k = \omega\sqrt{\mu\epsilon}$	wave number (rad/m)	Cutoff wave number
a	radius (m); TE: $p = p'_{nm}$ (TE_{11}: 1.841), TM: $p = p_{nm}$ (TM_{01}: 2.405)	Circular guide cutoff
f	frequency (Hz)	Ideal reactances
L_s	lead inductance (H); inductive above f_r	Capacitor self-resonance
C_s	winding capacitance (F); capacitive above f_r	Inductor self-resonance
V_1, V_2	input and output voltages	High-pass gain
a_n, b_n	incident and outgoing waves at port n ($\sqrt{\text{W}}$)	Power waves
b_s	source wave	Flow graph with the source
Γ_S	source reflection coefficient	Flow graph with the source
d_{gs}	centre in the Γ_S plane	Gain-circle centre
r_{gs}	radius	Gain-circle radius
U	radiation intensity (W/sr)	Directivity
P_{rad}	radiated power (W)	Directivity
θ_{HP}, ϕ_{HP}	half-power beamwidths (deg)	Directivity estimate
k	radiation efficiency	Gain
P_n	normalized power pattern; sr	Beam area
V_{oc}	open-circuit voltage (V)	Effective height

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Symbol	Meaning	First used in
I_0	terminal current (A); half-wave dipole λ/π	Effective height
$Z_0 = 377\ \Omega$; R_r	radiation resistance (Ω)	Height and aperture
P_r	received power (W)	Effective aperture
A_p	physical aperture (m^2)	Effective aperture
λ	wavelength (m)	Aperture and directivity
$\sin\theta\, d\theta\, d\phi$	element of solid angle (sr)	Radiated power
δ	progressive feed phase (rad)	Phase difference
N	number of elements	Array factor
D	largest antenna size (m); far field beyond r_2	Field regions
e	emissivity	Brightness temperature
T_{phys}	physical temperature (K)	Brightness temperature
T_{sys}	system noise temperature (K)	Radiometer resolution
τ	integration time (s)	Radiometer resolution
P_t	transmitted power (W)	Density at the target
G	antenna gain	Density at the target
R	range (m)	Density at the target
σ	radar cross-section (m^2)	Radar equation
P_{min}	minimum detectable signal (W)	Maximum range
t	echo delay (s)	Range from delay
σ_t	tissue conductivity (S/m)	Specific absorption rate
ρ	tissue density (kg/m^3); W/kg	Specific absorption rate
S_{lim}	power-density limit (W/m^2)	Safe distance

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